



Time-varying disturbance-observer-based tracking control of uncertain flexible-joint manipulator

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ABSTRACT

This paper presents a time-varying disturbance-observer- (TVDO-) based improved sliding-mode control (ISMC) method to achieve tracking control of uncertain flexible-joint manipulator (UFJM). Based on the UFJM dynamic model and its ideal dynamic model regardless of uncertainties, an error system in the form of an integral chain with disturbances is established. The tracking control objective of the UFJM is achieved by designing the TVDO-based ISMC law to guarantee the ultimate boundedness of link angle errors for the error system. Specifically, by employing the TVDO, the disturbances in the error system are estimated. By using the TVDO-based ISMC method, the uncertainties are overcome, and the chattering problem is reduced. The contradiction between high tracking accuracy and suitable torque without peaking phenomenon is alleviated. Finally, the stability analysis of the closed-loop control system is presented, and the validity of the proposed method is illustrated by simulation and experiment.

1. Introduction

In recent decades, some research on tracking control of the flexible-joint manipulator (FJM) has been attracting considerable attention (Ding et al., 2022; Luca & Book, 2008; Spyarakos-Papastavridis & Dai, 2021). This type of manipulator uses some joints made of low-stiffness or elastic materials. As a result, they show flexible features and remarkable advantages in ensuring human-machine safety (Luca & Book, 2008). Compared to a conventional rigid-joint manipulator, the FJM has more complex physical characteristics. For a rigid-joint manipulator, the actuator is rigidly connected to the link, so its actuator angle ideally coincides with the coaxial link angle, and its dynamic model has the same number of control inputs as the degrees of freedom. However, for the FJM, due to the effect of joint flexibility (Liu et al., 2019), the actuator angle and the coaxial link angle revolving in the same link are usually not equal, the link angle cannot be controlled directly by the actuator torque, and the manipulator is prone to vibration (Nanos & Papadopoulos, 2015). In other words, the numbers of independent control inputs for the FJM are less than its degrees of freedom, so the FJM has underactuated characteristics (Meng et al., 2022; Urbas et al., 2020). Therefore, universal FJMs other than planar single-link ones are not feedback linearizable by static state feedback (Luca & Book, 2008). Moreover, control of the FJM in practical application is also affected

by some model uncertainties and disturbance uncertainties (Jin et al., 2017) (these are collectively called “uncertainties” for convenience). Therefore, tracking control of the uncertain flexible-joint manipulator (UFJM) has enormous challenges.

In order to achieve the tracking control of the UFJM, in Yan et al. (2021), a robust control method was employed, where an equivalent input disturbance was proposed to compensate for uncertainties. In Hu et al. (2021), a dynamic surface control method was utilized based on a nonlinear disturbance observer (DO) and an interval type-2 fuzzy neural network approximator (IT2NNA) to achieve tracking control. In addition, some advanced control techniques, such as time-scale expansion-based control (Zhang & Li, 2017), adaptive backstepping control (Dian et al., 2019), sliding-mode control (Rsetam et al., 2020), and distributed higher-order differential feedback control (John & Zafer, 2017), were also presented for the UFJM. However, the control object of these above methods is the UFJM with a single link, and these methods can hardly be applied or extended to a universal UFJM.

To tackle the tracking control problem of a universal UFJM, Ruderman et al. designed a virtual torsion sensor to compensate for flexible-joint torsion, and achieved tracking control by employing a proportional-derivative controller (Ruderman & Iwasaki, 2016). However, this method easily leads to overshoot. Yu et al. designed an

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adaptive output feedback control method, which can be applied to the case that the UFJM rotates in a confined space (Yu, He, Li, & Sun, 2021). Kelekci et al. presented a model-free technique based on the interval type-2 fuzzy logic method (Kelekci & Kizir, 2019), but the lack of a dynamic model makes the method difficult to attain good tracking performance. Wang et al. transformed a universal UFJM into a flatness dynamic model with a complex lumped disturbance and then designed a finite-time DO-based output feedback controller (Wang, Zhang, et al., 2021). Besides, some people employed the minimally model-based variable impedance controller (Spyrakos-Papastavridis & Dai, 2021), adaptive impedance controller based on Bayesian estimation (Yu, He, Li, Xue, et al., 2021), passivity-based controller (Keppler et al., 2016), adaptive fuzzy controllers (Ling et al., 2020; Sun et al., 2019), etc. for the UFJM. Although a lot of work has been done, the tracking control accuracy of the FJM is usually degraded by the effect of uncertainties. On the other hand, most of the existing methods for this system do not consider the torque amplitude, which usually results in a large destructive torque for the system. Due to the underactuated characteristics and complex nonlinearity of the system, it is challenging and urgent work to achieve high-precision control of the uncertain system with suitable torque. Hence, this work is of great importance in the application of the manipulator.

Among diverse control methods, the sliding-mode control (SMC) method has a powerful ability to overcome uncertainties for nonlinear systems, but it is well known that this method has the disadvantage of chattering (Cong et al., 2012; Liao et al., 2019). For the UFJM, this problem is worsened because the common SMC law usually requires a large switching amplitude to suppress disturbances related to states and ensure robustness (Ginoya et al., 2014; Wang, Zhang, et al., 2021). Some methods have been developed to deal with the chattering problem. Since a super-twisting (sliding-mode) algorithm generates a continuous control signal that reduces chattering, Liao et al. designed a damping controller based on this and used time-varying parameters in the algorithm to compensate for disturbances with states (Liao et al., 2019). However, this method is applicable to systems with relative degree one (Liao et al., 2019), whereas the UFJM is a high-relative-degree system (Samiei et al., 2022). Cong et al. used a DO to compensate for disturbances with states, incorporated an exponential time-varying function into a sliding-mode surface to shorten the reaching phase and then designed a DO-based SMC method to reduce the large chattering (Cong et al., 2012). However, since the UFJM has underactuated characteristics, when we use these SMC methods for the UFJM, it is hard to stabilize both actuated variables and underactuated variables in the sliding-mode surface simultaneously (Zaare et al., 2019).

To achieve sliding-mode tracking control of the UFJM, Xiong et al. transformed the tracking control of the UFJM to direct SMC of a linear system by using a decentralized joint torque design. But the direct SMC method does not consider mismatched disturbances in the UFJM model (Xiong et al., 2019). A double-loop control method can be applied to the UFJM, in which a sliding-mode-based controller with an extended state observer was designed to ensure only the convergence of the underactuated variables in the outer loop (Wang, Liu, et al., 2021). Although this method can achieve tracking control of the UFJM, it did not consider potential peaking phenomenon of torque. Moreover, from the perspective of promoting the development of control theory, it is still valuable to study how to control both actuated and underactuated variables simultaneously by using only one observer-based SMC method.

Taking these problems into account, some people enhanced the DO-based SMC method by adding estimated disturbance values from a DO to the sliding-mode surface, so as to achieve tracking control of the UFJM (Alam et al., 2019; Ginoya et al., 2014; Nam et al., 2018). However, these three methods were not applicable to a universal UFJM. Moreover, when the DO is required to estimate uncertainties more accurately by adjusting design parameters, its estimation errors easily have huge values at the initial period (José et al., 2009; Susmann &

Kokotovic, 1991), which leads to the peaking phenomenon of torque. This problem is also prone to occur in some active disturbance rejection control methods with constant-gain extended state observers (Rsetam et al., 2020; Wang, Liu, et al., 2021). Therefore, it is necessary to design a new DO-based SMC method that can not only be applied to a general UFJM but also alleviate the contradiction between high tracking accuracy and suitable torque.

In order to achieve tracking control of the UFJM with high tracking accuracy and suitable torque, in this paper, we present a time-varying disturbance-observer- (TVDO-) based improved sliding-mode control (ISMC) method. Since the existence of uncertainties makes a distinction between the outputs of the UFJM dynamic model and the outputs of the ideal dynamic FJM model, we use this point to establish an error system. The main body of the error system is written in the form of an integral chain. The rest parts of the error system contain uncertainties of the UFJM dynamic model, ideal model state variables, and error system state variables. We consider these as matched and mismatched disturbances of the main body in the form of an integral chain. Then, the tracking control problem for the UFJM is transformed to design control torque to guarantee the ultimate boundedness of link angle errors for the error system.

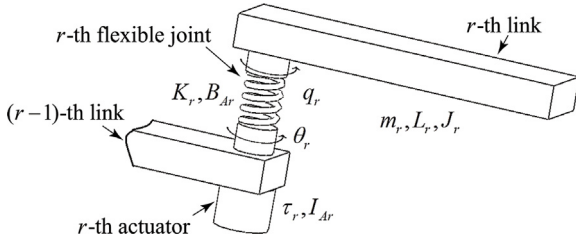
Considering that the matched and mismatched disturbances in the error system may have large upper boundary values and fluctuate over a wide range as the manipulators move, here, we design a TVDO to estimate the disturbances. Then, we propose a TVDO-based ISMC method, in which the ISMC law includes time-varying design parameters. In this way, the proposed method achieves tracking control of the UFJM, overcomes uncertainties, reduces the chattering problem, and alleviates the contradiction between high tracking accuracy and suitable torque without peaking phenomenon. Finally, the stability of the closed-loop control system is proved, and the validity of the proposed method is illustrated by simulation results and experimental results.

The novelties and contributions of this paper are summarized as follows.

- A novel DO with time-varying design parameters is designed, which provides an opportunity to improve both transient-state and steady-state control performances for the TVDO-based control system by designing time-varying design parameters.
- The SMC method is improved in this paper, and the ISMC method not only handles the matched and mismatched disturbances but also effectively improves the tracking accuracy of the UFJM.
- The TVDO-based ISMC method can be applied to a universal UFJM. Differing from common DO-based SMC methods, this method alleviates the contradiction between high tracking accuracy and suitable torque without peaking phenomenon for the UFJM. Besides, we provide users with a rational way to select its design parameters according to desired specifications.

The rest of this paper is organized as follows. Section 2 presents the dynamic model and corresponding error system of the UFJM. Section 3 designs a TVDO to estimate disturbances in the error system. Based on the TVDO, Section 4 proposes an ISMC method to achieve the tracking control objective of the UFJM. Finally, Section 5 gives simulations and an experiment for validity verification of this method.

Notations. To avoid unnecessary confusion, we make some mathematical illustrations for the rest of this paper. The differentiation and derivation of matrix functions are obtained based on the numerator layout. $y_1 = \exp(\diamond)$ denotes the exponential function of $\diamond$, $y_2 = \text{sat}(\diamond)$ denotes the saturation function of $\diamond$, $y_3 = \text{sgn}(\diamond)$ denotes the sign function of $\diamond$, t is the time, and $\|\Psi\|$ denotes the norm of the matrix Ψ . Let $a \geq 2$ and b_2 denote integers. Then, $b_1 E b_2$ denotes $b_1 \times 10^{b_2}$, $I_a \in \mathbb{R}^{a \times a}$ denotes an identity matrix, $O_a \in \mathbb{R}^{a \times a}$ denotes a zero matrix, and $\text{diag}(\ddagger_1, \ddagger_2, \dots, \ddagger_a)$ denotes a block diagonal matrix whose principal diagonal elements are $\ddagger_1, \ddagger_2, \dots, \ddagger_a$. For a matrix $Y \in \mathbb{R}^{a \times a}$, and $Y^a = \underbrace{Y Y \dots Y}_a$.

Fig. 1. Space model of the r th flexible joint ($r \geq 2$).

2. Model description and error system establishment

In this section, the UFJM dynamic model and its ideal dynamic model are described. Based on these two models, an error system in the form of an integral chain is established. Then, how to achieve the tracking control objective of the UFJM is discussed based on the error system.

2.1. Model description

For the UFJM, each joint of this manipulator is flexible. Let q_r , θ_r , and τ_r be the link angle, the actuator angle, and the actuator torque of the r th link, respectively ($r = 1, 2, \dots, n$). The space model of the r th ($r \geq 2$) flexible joint is depicted in Fig. 1, in which m_r and L_r are the mass and the length of the r th link, J_r is the inertia about link shaft, I_{Ar} is the inertia of the actuator, K_r denotes the flexible-joint stiffness, and B_{Ar} denotes the damping.

By using the Euler–Lagrange dynamic equations, the dynamic model of the UFJM is described as the following form (Ghorbel et al., 1989; Sun et al., 2019):

$$\begin{aligned} M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + K \cdot (q - \theta) &= d_1(p, t) \\ I_A\ddot{\theta} - K \cdot (q - \theta) + B_A\dot{\theta} &= \tau + d_2(p, t) \end{aligned} \quad (1)$$

where $q := [q_1, q_2, \dots, q_n]^T$, $\theta := [\theta_1, \theta_2, \dots, \theta_n]^T$, $\tau := [\tau_1, \tau_2, \dots, \tau_n]^T$, $M(q) \in \mathbb{R}^{n \times n}$ is an inertia matrix, $C(q, \dot{q}) \in \mathbb{R}^n$ is a vector containing centripetal forces and Coriolis forces, $G(q) \in \mathbb{R}^n$ is a gravity vector, $K \in \mathbb{R}^{n \times n}$ represents the joint flexibility, $I_A \in \mathbb{R}^{n \times n}$ is an actuator inertia matrix, $B_A \in \mathbb{R}^{n \times n}$ is a vector of natural damping, $p \in \mathbb{R}^{4n}$ is the set of q , $\dot{q}$, θ and $\dot{\theta}$, $p := [q^T, \dot{q}^T, \theta^T, \dot{\theta}^T]^T$, and $d_i(p, t) \in \mathbb{R}^n$ ($i = 1, 2$), which is abbreviated to d_i , is a vector of unknown uncertainties. Besides, $d_i(p, t) := d_{Mi}(p, t) + d_{Di}(t)$, where $d_{Mi}(p, t) \in \mathbb{R}^n$ denotes model uncertainties vector caused by parameter perturbation and $d_{Di}(t) \in \mathbb{R}^n$ denotes external disturbance uncertainties vector. Among them, $\|d^j d_{Di}(t)/dt^j\|$ ($i = 0, 1, 2, 3$) is bounded, which has been commonly assumed in some control methods of practical mechanical systems (Ginoya et al., 2014; Li et al., 2019).

The dynamic model (1) has the following common properties (Cheng et al., 2021): $M(q)$ is positive definite and symmetric, and both $M(q)$ and $M^{-1}(q)$ are uniformly bounded. Meanwhile, $\|C(q, \dot{q})\| \leq C_M \|\dot{q}\|$ and $\|G(q)\| \leq G_M$, where C_M and G_M are positive constants.

2.2. Error system establishment

We express a target angle of the r th link as q_{dr} ($r = 1, 2, \dots, n$) and define the target angles vector as $q_d := [q_{d1}, q_{d2}, \dots, q_{dn}]^T$. The tracking control objective is to design the control torque τ_r to make q converge to q_d as accurately as possible. Here, we first establish an error system in preparation for designing control law to achieve this objective.

The error system is established according to the dynamic model (1) and an ideal dynamic model of the FJM. For the ideal dynamic model, we define $\bar{q}_r$ and $\bar{\theta}_r$ as the r th ideal angles of the link and the actuator,

respectively, and define $\bar{\tau}_r$ as the r th ideal torque. Then, the ideal FJM dynamic model can be described as

$$\begin{aligned} \bar{M}(\bar{q})\ddot{\bar{q}} + \bar{C}(\bar{q}, \dot{\bar{q}})\dot{\bar{q}} + \bar{G}(\bar{q}) + K \cdot (\bar{q} - \bar{\theta}) &= 0 \\ I_A\ddot{\bar{\theta}} - K \cdot (\bar{q} - \bar{\theta}) + B_A\dot{\bar{\theta}} &= \bar{\tau} \end{aligned} \quad (2)$$

where $\bar{q} := [\bar{q}_1, \bar{q}_2, \dots, \bar{q}_n]^T$, $\bar{\theta} := [\bar{\theta}_1, \bar{\theta}_2, \dots, \bar{\theta}_n]^T$, $\bar{\tau} := [\bar{\tau}_1, \bar{\tau}_2, \dots, \bar{\tau}_n]^T$, and $\bar{M}(\bar{q})$, $\bar{C}(\bar{q}, \dot{\bar{q}})$ and $\bar{G}(\bar{q})$ are the nominal forms of $M(q)$, $C(q, \dot{q})$ and $G(q)$.

Since there exist uncertainties in the UFJM dynamic model (1), the states of (1) are apparently different from that of the ideal FJM dynamic model (2). We define the state errors vector $e_k \in \mathbb{R}^n$ ($k = 1, 2, 3, 4$) as $e_1 := q - \bar{q}$, $e_2 := \dot{q} - \dot{\bar{q}}$, $e_3 := \theta - \bar{\theta}$, $e_4 := \dot{\theta} - \dot{\bar{\theta}}$, and we define $e := [e_1^T, e_2^T, e_3^T, e_4^T]^T$, $e_k := [e_{k1}, e_{k2}, \dots, e_{kn}]^T$, where e_{kr} is the r th term of e_k ($r = 1, 2, \dots, n$). Based on (1) and (2), an error system in the form of an integral chain can be established:

$$\begin{cases} \dot{e}_1 = e_2, \dot{e}_2 = e_3 + \sigma_1 \\ \dot{e}_3 = e_4, \dot{e}_4 = f(e) + g\tau + \sigma_2 \end{cases} \quad (3)$$

where

$$\begin{aligned} g &= I_A^{-1}, f(e) = gK \cdot (e_1 - e_3) - gB_A e_4 \\ \sigma_1 &= D_1(e, t) - \ddot{\bar{q}} - e_3, \sigma_2 = D_2(e, t) - \ddot{\bar{\theta}} \end{aligned} \quad (4)$$

Here, $D_1(e, t) = M^{-1}(q)[-C(q, \dot{q})\dot{q} - G(q) - K(q - \theta) + d_1]$ $|_{q=e_1+\bar{q}, \dot{q}=e_2+\dot{\bar{q}}, \theta=e_3+\bar{\theta}}$, and $D_2(e, t) = g d_2$. Define $\sigma_i = [\sigma_{i1}, \sigma_{i2}, \dots, \sigma_{in}]^T$ ($i = 1, 2$). From (4), we find that σ_{ir} ($r = 1, 2, \dots, n$) is a complex unknown nonlinear term containing the uncertainties of the UFJM dynamic model, the states of the ideal dynamic model, and the states of the error system.

When we design control law based on the error system (3), two assumptions need to be given: (A1) The target angle q_{dr} from the vector q_d is supposed to be bounded, and its derivatives up to a necessary order 6 have Lipschitz constants; (A2) The actuator torques and the system energy are assumed to be bounded.

Then, we set $\bar{q} = q_d$, which implies that $e_1 = q - q_d$. Clearly, the tracking control objective for the UFJM is transformed to the control objective of designing τ to guarantee the ultimate boundedness of e_1 for the error system (3).

Remark 1. Assumption (A1) is reasonable because the target angle q_{dr} from the vector q_d is given in advance according to the practical requirements. Actually, q_{dr} under Assumption (A1) contains many types, such as constant, sine function, higher-order function, etc.

Remark 2. Limited by the manufacturing and materials of the system, Assumption (A2) is feasible and common in the control of practical-oriented uncertain mechanical systems (Jin et al., 2017; Mohammadi et al., 2013; Sun et al., 2019).

To achieve the control objective of the error system (3), on the one hand, we can obtain the states vector e of the error system (3). It is because that q_r , $\dot{q}_r$, θ_r , and $\dot{\theta}_r$ can be measured by employing encoders and tachometers, $\bar{q} = q_d$ is given, and then $\dot{\bar{q}}$ is easy to obtain, and $\bar{\theta}$, $\dot{\bar{\theta}}$ and $\ddot{\bar{\theta}}$ are calculated based on the ideal FJM dynamic model (2):

$$\begin{aligned} \ddot{\bar{\theta}} &= K^{-1}[\bar{M}\ddot{\bar{q}} + \bar{C}\dot{\bar{q}} + \bar{G} + K\bar{q}] \\ \ddot{\bar{q}} &= K^{-1}[\bar{M}\ddot{\bar{q}} + \bar{M}\ddot{\bar{q}}^{(3)} + \bar{C}\dot{\bar{q}} + \bar{C}\ddot{\bar{q}} + \bar{G} + K\dot{\bar{q}}] \\ \ddot{\bar{\theta}} &= K^{-1}[2\bar{M}\ddot{\bar{q}}^{(3)} + \bar{M}\ddot{\bar{q}} + \bar{M}\ddot{\bar{q}}^{(4)} + \bar{C}\ddot{\bar{q}} + 2\bar{C}\dot{\bar{q}} \\ &\quad + \bar{C}\ddot{\bar{q}}^{(3)} + \bar{G} + K\ddot{\bar{q}}] \end{aligned} \quad (5)$$

On the other hand, we consider σ_i in the error system (3) as a vector of disturbances. Since the error system (3) can be written as a system in the form of an integral chain with disturbances, we can design a time-varying disturbance observer (TVDO) to estimate σ_i . Then, we will design a TVDO-based improved sliding-mode control (ISMC) method to guarantee the ultimate boundedness of e_1 .

3. Time-varying disturbance-observer design

For the disturbances vector σ_i in the error system (3), because σ_i is influenced by many different variables, it fluctuates over a wide range. Hence, a general SMC law for the error system (3) with disturbances needs a larger switching amplitude to suppress σ_i in a wide range and then makes the system exhibit large chattering in the steady state.

To alleviate chattering of the system (3), an effective improvement method is to add the estimated vector of σ_1 , in the integral chain of the system (3), into the sliding-mode surface and design the corresponding SMC law with estimated vectors of σ_1 , σ_2 , and the derivative $\dot{\sigma}_1$. In this way, this method requires a smaller switching amplitude to mainly suppress disturbance estimation errors. For the estimation of σ_1 , σ_2 , and specially $\dot{\sigma}_1$, the disturbance observer (DO) can fulfill the requirement. However, using the method to achieve high-accuracy tracking control is usually accompanied by the peaking phenomenon of torque. To alleviate this contradiction, we propose a time-varying disturbance observer (TVDO).

Let the estimated vectors of σ_i , $\dot{\sigma}_i$, and $\ddot{\sigma}_i$ be $\hat{\sigma}_i$, $\dot{\hat{\sigma}}_i$, and $\ddot{\hat{\sigma}}_i$, respectively ($i = 1, 2$). The TVDO is designed as

$$\begin{cases} \hat{\sigma}_i := [\hat{\sigma}_{i1}, \hat{\sigma}_{i2}, \dots, \hat{\sigma}_{in}]^T := \gamma_{i1} + l_{i1} h_i(t) e_w \\ \dot{\hat{\sigma}}_i := [\dot{\hat{\sigma}}_{i1}, \dot{\hat{\sigma}}_{i2}, \dots, \dot{\hat{\sigma}}_{in}]^T := \gamma_{i2} + l_{i2} h_i^2(t) e_w \\ \ddot{\hat{\sigma}}_i := [\ddot{\hat{\sigma}}_{i1}, \ddot{\hat{\sigma}}_{i2}, \dots, \ddot{\hat{\sigma}}_{in}]^T := \gamma_{i3} + l_{i3} h_i^3(t) e_w \end{cases} \quad (6)$$

where $l_{iv} \in \mathbb{R}^n$ and $h_i(t) \in \mathbb{R}^n$ are selective design matrices, $l_{iv} := \text{diag}(l_{iv1}, l_{iv2}, \dots, l_{ivn})$, $l_{iv} > 0$ is a constant ($i = 1, 2$; $v = 1, 2, 3$; $r = 1, 2, \dots, n$), e_w represents the state errors vector e_2 or e_4 ($w = 2i$), and $\gamma_{iv} \in \mathbb{R}^n$ is an auxiliary vector such that

$$\begin{cases} \dot{\gamma}_{11} := -l_{11}[h_1(t)(e_3 + \hat{\sigma}_1) + \dot{h}_1(t)e_2] + \dot{\hat{\sigma}}_1 \\ \dot{\gamma}_{12} := -l_{12}[h_1^2(t)(e_3 + \hat{\sigma}_1) + 2h_1(t)\dot{h}_1(t)e_2] + \dot{\hat{\sigma}}_1 \\ \dot{\gamma}_{13} := -l_{13}[h_1^3(t)(e_3 + \hat{\sigma}_1) + 3h_1^2(t)\dot{h}_1(t)e_2] \\ \dot{\gamma}_{21} := -l_{21}[h_2(t)(f(e) + g\tau + \hat{\sigma}_2) + \dot{h}_2(t)e_4] + \dot{\hat{\sigma}}_2 \\ \dot{\gamma}_{22} := -l_{22}[h_2^2(t)(f(e) + g\tau + \hat{\sigma}_2) + 2h_2(t)\dot{h}_2(t)e_4] + \dot{\hat{\sigma}}_2 \\ \dot{\gamma}_{23} := -l_{23}[h_2^3(t)(f(e) + g\tau + \hat{\sigma}_2) + 3h_2^2(t)\dot{h}_2(t)e_4] \end{cases} \quad (7)$$

Here, to alleviate the contradiction between high tracking accuracy and suitable torque without peaking phenomenon, we first define $h_i(t)$ as a time-varying design matrix:

$$h_i(t) := \hat{h}_i(t) I_n \quad (8)$$

where $\hat{h}_i(t)$ is a scalar function. Clearly, the selective design matrix $h_i(t)$ is determined by $\hat{h}_i(t)$, and $\hat{h}_i(t)$ needs to be selected under the following conditions: (CA1) $\hat{h}_i(t)$ varies slowly at the initial period, $1 \leq \hat{h}_i(t) \leq \mu_2$, and the constant design parameter $\mu_2 \geq 1$; (CA2) The first derivative of $\hat{h}_i(t)$ is continuous and bounded, and given an arrival moment t_1 ($0 < t_1 < \infty$) at the initial period of system response: when $t < t_1$, $\hat{h}_i(t) \leq 0$, and when $t \geq t_1$, $\hat{h}_i(t) = 0$ and $\dot{\hat{h}}_i(t) = \mu_3$ (i.e., $\hat{h}_i(t)$ arrives μ_3), in which $\mu_3 \geq 1$ is a constant design parameter. Under Conditions (CA1) - (CA2), the trajectory of $\hat{h}_i(t)$ should be a trajectory that when $t \leq t_1$, the trajectory of $\hat{h}_i(t)$ is a straight line segment or a curve segment of continuous or intermittent decline, and when $t > t_1$, the trajectory of $\hat{h}_i(t)$ is a straight line.

Taking the derivatives of $\hat{\sigma}_i$, $\dot{\hat{\sigma}}_i$ and $\ddot{\hat{\sigma}}_i$ in (6) yields

$$\begin{cases} \dot{\hat{\sigma}}_i = \dot{\gamma}_{i1} + l_{i1}[h_i(t)(e_3 + \sigma_1) + \dot{h}_i(t)e_2] \\ \dot{\dot{\hat{\sigma}}}_i = \dot{\gamma}_{i2} + l_{i2}[h_i^2(t)(e_3 + \sigma_1) + 2h_i(t)\dot{h}_i(t)e_2] \\ \dot{\ddot{\hat{\sigma}}}_i = \dot{\gamma}_{i3} + l_{i3}[h_i^3(t)(e_3 + \sigma_1) + 3h_i^2(t)\dot{h}_i(t)e_2] \\ \dot{\hat{\sigma}}_2 = \dot{\gamma}_{21} + l_{21}[h_2(t)(f(e) + g\tau + \sigma_2) + \dot{h}_2(t)e_4] \\ \dot{\dot{\hat{\sigma}}}_2 = \dot{\gamma}_{22} + l_{22}[h_2^2(t)(f(e) + g\tau + \sigma_2) + 2h_2(t)\dot{h}_2(t)e_4] \\ \dot{\ddot{\hat{\sigma}}}_2 = \dot{\gamma}_{23} + l_{23}[h_2^3(t)(f(e) + g\tau + \sigma_2) + 3h_2^2(t)\dot{h}_2(t)e_4] \end{cases} \quad (9)$$

Note that $\dot{\hat{\sigma}}_i$ and $\dot{\sigma}_i$ as well as $\ddot{\hat{\sigma}}_i$ and $\ddot{\sigma}_i$ are different concepts. Substituting $\dot{\gamma}_{iv}$ of (7) into (9) yields

$$\begin{aligned} \dot{\hat{\sigma}}_i &= l_{i1} h_i(t)(\sigma_i - \hat{\sigma}_i) + \dot{\hat{\sigma}}_i, \quad \dot{\dot{\hat{\sigma}}}_i = l_{i2} h_i^2(t)(\sigma_i - \hat{\sigma}_i) + \dot{\dot{\hat{\sigma}}}_i \\ \dot{\ddot{\hat{\sigma}}}_i &= l_{i3} h_i^3(t)(\sigma_i - \hat{\sigma}_i) \end{aligned} \quad (10)$$

We define the errors vector of disturbances estimation as

$$z_i := [\tilde{\sigma}_i^T, \tilde{\dot{\sigma}}_i^T, \tilde{\ddot{\sigma}}_i^T]^T := [(\sigma_i - \hat{\sigma}_i)^T, (\dot{\sigma}_i - \dot{\hat{\sigma}}_i)^T, (\ddot{\sigma}_i - \ddot{\hat{\sigma}}_i)^T]^T \quad (11)$$

where $\tilde{\sigma}_i := [\tilde{\sigma}_{i1}, \tilde{\sigma}_{i2}, \dots, \tilde{\sigma}_{in}]^T$, $\tilde{\dot{\sigma}}_i := [\tilde{\dot{\sigma}}_{i1}, \tilde{\dot{\sigma}}_{i2}, \dots, \tilde{\dot{\sigma}}_{in}]^T$, and $\tilde{\ddot{\sigma}}_i := [\tilde{\ddot{\sigma}}_{i1}, \tilde{\ddot{\sigma}}_{i2}, \dots, \tilde{\ddot{\sigma}}_{in}]^T$. Subtracting $\dot{\hat{\sigma}}_i$, $\dot{\dot{\hat{\sigma}}}_i$, $\dot{\ddot{\hat{\sigma}}}_i$ of (10) from $\dot{\sigma}_i$, $\dot{\dot{\sigma}}_i$, $\dot{\ddot{\sigma}}_i$ respectively yields

$$\begin{aligned} \dot{\tilde{\sigma}}_i &= -l_{i1} h_i(t) \tilde{\sigma}_i + \tilde{\dot{\sigma}}_i, \quad \dot{\tilde{\dot{\sigma}}}_i = -l_{i2} h_i^2(t) \tilde{\sigma}_i + \tilde{\dot{\dot{\sigma}}}_i \\ \dot{\tilde{\ddot{\sigma}}}_i &= -l_{i3} h_i^3(t) \tilde{\sigma}_i + \tilde{\dot{\ddot{\sigma}}}_i \end{aligned} \quad (12)$$

where $\tilde{\dot{\sigma}}_i$, $\tilde{\dot{\dot{\sigma}}}_i$, and $\tilde{\dot{\ddot{\sigma}}}_i$ are the derivatives of $\tilde{\sigma}_i$, $\tilde{\dot{\sigma}}_i$, and $\tilde{\ddot{\sigma}}_i$.

According to (12), let

$$L_{Ai} := \begin{bmatrix} -l_{i1} & I_n & O_n \\ -l_{i2} & O_n & I_n \\ -l_{i3} & O_n & O_n \end{bmatrix}, \quad L_{Bi} := \begin{bmatrix} -l_{i1} h_i(t) & I_n & O_n \\ -l_{i2} h_i^2(t) & O_n & I_n \\ -l_{i3} h_i^3(t) & O_n & O_n \end{bmatrix} \quad (13)$$

Here, we set all the eigenvalues of L_{Ai} in the left half plane by selecting suitable l_{iv} , thereby the Lyapunov equation

$$L_{Ai}^T P_i + P_i L_{Ai} = -I_{3n}, \quad i = 1, 2 \quad (14)$$

always has unique positive definite solution $P_i = P_i^T \in \mathbb{R}^{3n \times 3n}$.

Let $\Lambda_\theta(P_i)$ be the θ -th eigenvalue of P_i ($\theta = 1, 2, \dots, 3n$), and let $\Lambda_{\min}(P_i)$ and $\Lambda_{\max}(P_i)$ be the minimal and maximal eigenvalues of P_i .

Theorem 1. Under Assumptions (A1) and (A2), if the TVDO (6) is employed, the errors vector of disturbances estimation, z_i , is ultimately bounded, and its ultimate boundary value can be adjusted to be small enough after a sufficiently long time by selecting a larger μ_3 and setting l_{iv} to make L_{Ai} in (13) produce eigenvalues with larger negative real parts.

Proof. Define

$$\begin{aligned} \eta_{i1} &:= h_i^3(t) \tilde{\sigma}_i, \quad \eta_{i2} := h_i^2(t) \tilde{\dot{\sigma}}_i, \quad \eta_{i3} := h_i(t) \tilde{\ddot{\sigma}}_i \\ \eta_i &:= [\eta_{i1}^T, \eta_{i2}^T, \eta_{i3}^T]^T \end{aligned} \quad (15)$$

According to (8), (12) and (15), we have

$$\begin{cases} \dot{\eta}_{i1} = h_i(t)(\eta_{i2} - l_{i1} \eta_{i1}) + 3\dot{h}_i(t) h_i^{-1}(t) \eta_{i1} \\ \dot{\eta}_{i2} = h_i(t)(\eta_{i3} - l_{i2} \eta_{i1}) + 2\dot{h}_i(t) h_i^{-1}(t) \eta_{i2} \\ \dot{\eta}_{i3} = h_i(t)(\sigma_i^{(3)} - l_{i3} \eta_{i1}) + \dot{h}_i(t) h_i^{-1}(t) \eta_{i3} \end{cases} \quad (16)$$

Next, we define the following Lyapunov function:

$$\begin{aligned} V_A(\eta_i) &= \eta_i^T P_i \eta_i \\ \Lambda_{\min}(P_i) \|\eta_i\|^2 &\leq V(\eta_i) \leq \Lambda_{\max}(P_i) \|\eta_i\|^2 \end{aligned} \quad (17)$$

Combining (8) and (16) and deriving $V_A(\eta_i)$ along η_{iv} yield

$$\begin{aligned} \dot{V}_A(\eta_i) &= \sum_{v=1}^3 \frac{\partial V_A(\eta_i)}{\partial \eta_{iv}} \dot{\eta}_{iv} \\ &= \dot{h}_i(t) \left[\frac{\partial V_A(\eta_i)}{\partial \eta_{i1}} (\eta_{i2} - l_{i1} \eta_{i1}) + \frac{\partial V_A(\eta_i)}{\partial \eta_{i2}} (\eta_{i3} - l_{i2} \eta_{i1}) - \frac{\partial V_A(\eta_i)}{\partial \eta_{i3}} \right. \\ &\quad \left. \cdot l_{i3} \eta_{i1} \right] + \dot{h}_i(t) \frac{\partial V_A(\eta_i)}{\partial \eta_{i3}} \sigma_i^{(3)} + \sum_{v=1}^3 \frac{(3-v+1)\dot{h}_i(t)}{h_i(t)} \frac{\partial V_A(\eta_i)}{\partial \eta_i} \eta_{iv} \end{aligned} \quad (18)$$

For (18), based on (14) and (17), there exists

$$\begin{aligned} &\frac{\partial V_A(\eta_i)}{\partial \eta_{i1}} (\eta_{i2} - l_{i1} \eta_{i1}) - \frac{\partial V_A(\eta_i)}{\partial \eta_{i3}} l_{i3} \eta_{i1} + \frac{\partial V_A(\eta_i)}{\partial \eta_{i2}} (\eta_{i3} - l_{i2} \eta_{i1}) \\ &= \eta_i^T (L_{Ai}^T P_i + P_i L_{Ai}) \eta_i = -\eta_i^T \eta_i = -\|\eta_i\|^2 \leq 0 \\ &\left| \frac{\partial V_A(\eta_i)}{\partial \eta_{iv}} \right| \leq \left| \frac{\partial V_A(\eta_i)}{\partial \eta_i} \right| = \|2\eta_i^T P_i\| \leq 2\Lambda_{\max}(P_i) \|\eta_i\| \end{aligned} \quad (19)$$

Combining the dynamic model (1), the assumptions of the system, and (5), we obtain that σ_i of (4) and its j th derivatives are bounded ($i = 1, 2$;

$j = 0, 1, 2, 3$). It implies that $\|d^j \sigma_i / dt^j\| \leq \mu_1$, where μ_1 is the unknown boundary and $0 < \mu_1 < \infty$. On this basis, substituting (19) into (18) yields

$$\begin{aligned} \dot{V}_A(\eta_i) &\leq -\dot{h}_i(t) \|\eta_i\|^2 + \left(\mu_1 \dot{h}_i(t) + \frac{6\dot{h}_i(t)}{h_i(t)} \|\eta_i\| \right) \left| \frac{\partial V_A(\eta_i)}{\partial \eta_{iv}} \right| \\ &\leq -\|\eta_i\| \left[\left(\dot{h}_i(t) - 12\Lambda_{\max}(P_i) \frac{\dot{h}_i(t)}{h_i(t)} \right) \|\eta_i\| - 2\mu_1 \dot{h}_i(t) \Lambda_{\max}(P_i) \right] \end{aligned} \quad (20)$$

Thus, after a sufficiently long time, we obtain that

$$\|\eta_i\| \leq \chi_{A0}, \quad \chi_{A0} = \frac{2\mu_1 \dot{h}_i(t) \Lambda_{\max}(P_i)}{\dot{h}_i(t) - 12\Lambda_{\max}(P_i) \dot{h}_i(t)/h_i(t)} \quad (21)$$

According to Conditions (CA1) - (CA2) that $\dot{h}_i(t) \geq 1$ and $\dot{h}_i(t) \leq 0$, we obtain that $-12\Lambda_{\max}(P_i) \dot{h}_i(t)/h_i(t) \geq 0$. Then, $\dot{h}_i(t) - 12\Lambda_{\max}(P_i) \dot{h}_i(t)/h_i(t) \geq 1$, which implies $0 < \chi_{A0} < \infty$. Thus, after a sufficiently long time, η_i is ultimately bounded.

Then, based on (8), (11), (15) and (21), since $\dot{h}_i^3(t) \geq \dot{h}_i^2(t) \geq \dot{h}_i(t) \geq 1$, after a sufficiently long time that $t \geq T_A$ ($0 < T_A < \infty$), $\|z_i\|$ is bounded by

$$\|z_i\| \leq \frac{\|\eta_i\|}{\dot{h}_i(t)} \leq \chi_{A1}, \quad \chi_{A1} = \frac{2\mu_1 \Lambda_{\max}(P_i)}{\dot{h}_i(t) - 12\Lambda_{\max}(P_i) \dot{h}_i(t)/h_i(t)} \quad (22)$$

and the boundary value χ_{A1} varies in pace with $\dot{h}_i(t)$. T_A is the unknown convergent time of the estimated values in the TVDO. This indicates that z_i is ultimately bounded.

From Condition (CA2), we know when $t \geq t_1$, $\dot{h}_i(t) = \mu_3 \geq 1$ and $\dot{h}_i(t)/h_i(t) = 0$. Hence, based on (22), after a sufficiently long time, the ultimate boundary of $\|z_i\|$ specifically narrows to

$$\|z_i\| \leq \chi_{A2}, \quad \chi_{A2} = 2\mu_1 \Lambda_{\max}(P_i) / \mu_3 \quad (23)$$

Thus, the boundary values χ_{A1} and χ_{A2} can be reduced by selecting larger μ_3 and smaller $\Lambda_{\max}(P_i)$. From (14), to make $\Lambda_{\max}(P_i)$ small enough, l_{iv} needs to be set to make L_{Ai} produce eigenvalues with larger negative real parts. ■

Therefore, from (11) and the above proof that the TVDO (6) makes $\hat{\sigma}_i$ approach σ_i , whose estimation accuracy can be greatly improved by selecting suitable l_{iv} and $\dot{h}_i(t)$.

Remark 3. We give some selective principles for the time-varying design parameter $\dot{h}_i(t)$ and its extra design constants μ_2 and μ_3 . Based on Conditions (CA1) and (CA2), $\dot{h}_i(t)$ starts from μ_2 and varies slowly at the initial period. When L_{Bi} is Hurwitz, and all of its eigenvalues are close to the origin, $\hat{\sigma}_i$ is able to approach σ_i well, which easily affects transient control performance at the initial time and induces a chattering or peaking phenomenon of torque. To improve tracking accuracy, see (23), (36), (46), and (48), the final value of $\dot{h}_i(t)$, i.e., μ_3 ($\mu_3 \geq 1$), and $\Lambda_{\max}(P_i)$ should be chosen to be as small as possible. Based on (13), (14) and the above analysis, since L_{Bi} is influenced by $\Lambda_{\max}(P_i)$ and the initial value of $\dot{h}_i(t)$, i.e., μ_2 , $\Lambda_{\max}(P_i)$ can be chosen as a relatively small one and the time-varying $\dot{h}_i(t)$ is designed to start with a large μ_2 ($\mu_2 \geq \mu_3 \geq 1$), which avoids the chattering or peaking phenomenon of torque at the initial time, and slide to the smaller final value μ_3 to improve the tracking accuracy. Note that, at the initial time near $t = 0$, the possible peaking phenomenon of torque still needs to be further solved, which will be discussed in Section 4. Here, we provide a simple selective example for $\dot{h}_i(t)$ under Conditions (CA1) and (CA2). We choose the trajectory of $\dot{h}_i(t)$ to be the decreasing part of the cycloidal curve from v_{H1} ($=\mu_2$) to v_{H2} ($=\mu_3$) when $t < t_1$, and a constant v_{H2} when $t \geq t_1$. Specifically, $\dot{h}_i(t)$ is described as

$$\dot{h}_i(t) = \begin{cases} (v_{H1} - v_{H2}) \left(\frac{1}{2\pi} \sin \frac{2\pi}{t_1} t - \frac{t}{t_1} \right) + v_{H1}, & t < t_1 \\ v_{H2}, & t \geq t_1 \end{cases} \quad (24)$$

where v_{H1} and v_{H2} are constants, and $v_{H1} \geq v_{H2} \geq 1$. This trajectory of $\dot{h}_i(t)$ is depicted in Fig. 10 of Appendix.

Remark 4. Many observers, such as finite-time (FT) observers, high-order sliding-mode (HOSM) observers (Boukhobza & Barbot, 1998),

and TVDOs can also be used to estimate σ_i ($i = 1, 2$). For example, an FT full-order DO was designed in Gu et al. (2022), and an FT ESO was designed in Huang et al. (2022). However, the estimated time derivative of $\hat{\sigma}_i$ is easily affected by the state estimation errors and design parameters since it is in the final channels of these two types of observers. Some HOSM observers (Benallegue et al., 2008; Fridman et al., 2008) also belong to the FT observers. This observer needs to transform the system to a Brunovsky canonical form, and then, based on the derivation and reconstruction of the observer information, $\hat{\sigma}_1$ can be obtained. Some TVDOs with good transient performance were presented in Giap et al. (2023), Razmjooei et al. (2022). However, these methods have challenges in calculating the estimated time derivative of $\hat{\sigma}_1$, because the TVDO in Giap et al. (2023) and the HOSM observer need to determine its derivatives of switching functions (Baratieri & Pinheiro, 2016), and the TVDO in Razmjooei et al. (2022) is prone to singular value problems. Unlike the above observers, using the proposed observer can easily obtain the estimated time derivative of $\hat{\sigma}_1$ with high estimation accuracy.

4. Improved sliding-mode control based on time-varying disturbance observer

In this section, we present the improved sliding-mode control (ISMC) method based on TVDO for the error system (3) to guarantee the ultimate boundedness of e_1 . Meantime, the ways to further alleviate the contradiction between high tracking accuracy and suitable torque under the TVDO-based ISMC method are illustrated.

We define S_r^* as the improved sliding-mode surface of the r th link ($r = 1, 2, \dots, n$), and we define the set of S_r^* as

$$S^* := [S_1^*, S_2^*, \dots, S_n^*]^T := S - S_0(t) \quad (25)$$

where

$$\begin{aligned} S &:= [S_1, S_2, \dots, S_n]^T := c_1 e_1 + c_2 e_2 + c_3 e_3 + e_4 + c_3 \hat{\sigma}_1 + \hat{\sigma}_1 \\ S_0(t) &:= S(0) \exp(-\alpha_0 t) \end{aligned} \quad (26)$$

in which $c_v = \text{diag}(c_{v1}, c_{v2}, \dots, c_{vn})$, and c_{vr} and α_0 are selective positive constants ($v = 1, 2, 3$; $r = 1, 2, \dots, n$). It is clear that subtracting the vector $S_0(t)$ from S makes S_r^* start from 0, which further avoids the negative effect on the peak value of torque (cf. Remark 8).

Let

$$L_{Cr} := \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -c_{1r} & -c_{2r} & -c_{3r} \end{bmatrix} \quad (27)$$

We define the eigenvalue of L_{Cr} as $\Lambda_{vr}(L_{Cr})$ ($v = 1, 2, 3$, $r = 1, 2, \dots, n$). The selective c_{vr} needs to satisfy Condition (CB1): The real part of each $\Lambda_{vr}(L_{Cr})$ is negative so that L_{Cr} is Hurwitz, $\Lambda_{vr}(L_{Cr}) \neq \alpha_0$, and $c_{1r} - c_{2r}\alpha_0 + c_{3r}\alpha_0^2 - \alpha_0^3 \neq 0$.

Theorem 2. For the error system (3), under Assumptions (A1) and (A2), e_1 is ultimately bounded, when the TVDO in (6) is employed, the sliding-mode surface of the r th link is set as (25), and the ISMC law is designed as

$$\begin{aligned} \tau &:= -g^{-1} [f(e) + \hat{\sigma}_2 + c_1 e_2 + c_2 (e_3 + \hat{\sigma}_1) + c_3 e_4 \\ &\quad + \lambda_A(t) S^* + \lambda_B S_{AT}(S^*)] \end{aligned} \quad (28)$$

where $\lambda_A(t) \in \mathbb{R}^{n \times n}$ is designed as a selective time-varying matrix, $\lambda_B \in \mathbb{R}^{n \times n}$ is a selective constant matrix,

$$\begin{aligned} \lambda_A(t) &= \text{diag}(\lambda_{A1}(t), \lambda_{A2}(t), \dots, \lambda_{An}(t)) \\ \lambda_B &= \text{diag}(\lambda_{B1}, \lambda_{B2}, \dots, \lambda_{Bn}) \\ S_{AT}(S^*) &= [\text{sat}(S_1^*), \text{sat}(S_2^*), \dots, \text{sat}(S_n^*)]^T \\ \text{sat } S_r^* &= \begin{cases} \text{sgn } S_r^*, |S_r^*| > \varepsilon \\ S_r^* / \varepsilon, |S_r^*| \leq \varepsilon \end{cases}, \quad r = 1, 2, \dots, n \end{aligned} \quad (29)$$

λ_{Br} and ε are constant positive design parameters, $\lambda_{Ar}(t)$ is a time-varying design parameter, which needs to meet the following conditions: (CB2)

$\lambda_{Ar}(t)$ is continuous and bounded; (CB3) Given an arrival moment t_2 ($0 < t_2 < \infty$) at the initial period of the system response: when $t < t_2$, $\lambda_{Ar}(t)$ satisfies that $\lambda_{Ar}(0) = \mu_{\lambda 1r}$ and $\mu_{\lambda 1r} \leq \lambda_{Ar}(t) \leq \mu_{\lambda 2r}$, and when $t \geq t_2$, $\lambda_{Ar}(t)$ satisfies that $\lambda_{Ar}(t) = \mu_{\lambda 3r} \geq 1$ (i.e., $\lambda_{Ar}(t)$ arrives $\mu_{\lambda 3r}$), in which $\mu_{\lambda 1r}$, $\mu_{\lambda 2r}$, and $\mu_{\lambda 3r}$ are positive constants, and $\mu_{\lambda 1r} \leq \mu_{\lambda 3r} \leq \mu_{\lambda 2r}$ ($r = 1, 2, \dots, n$).

Remark 5. Since the actual uncertainty vectors d_1 and d_2 are unknown, we need to set each element of $d_1(0)$ and $d_2(0)$ to 0. From (3), (4), (5), and (26), when $|q_{dr}^{(j)}(0)|$ ($j = 0, 1, 2, 3$) is given as a larger value, $|\hat{\sigma}_1(0)|$, $|\hat{\sigma}_1(0)|$ and $|\dot{S}(0)|$ contain larger values. Then, based on (25), (26), and (28), once $S^*(0)$ does not include $S_0(t)$, the torque vector occurs peaking phenomenon at the initial period through S^* in (28). Since $-S_0(t)$ is added into (25), we obtain that $0 \leq |S^*(0)| \leq |S(0)|$. Consequently, including $-S_0(t)$ in (25) is helpful to reduce the effect on the initial values of $\hat{\sigma}_1(0)$ and $\hat{\sigma}_1(0)$ and to reduce the peak value of torque at the initial period. In addition, when $|S^*| > 0$ at the initial period, there is still the possibility that $\lambda_A(t)S^*$ in (27) induces the peaking phenomenon of torque. Therefore, it is necessary to further design the time-varying $\lambda_A(t)$ to avoid this situation. Besides, considering $S_0(t)$ in (25) avoids excessive dependence on the selection of $\lambda_A(t)$.

Remark 6. To help users select the time-varying design parameter $\lambda_{Ar}(t)$ under Conditions (CB2) and (CB3), we give a clear and simple example that has an outstanding effect on improving the control performance of the system. Here, we set the trajectory of $\lambda_{Ar}(t)$ to be the ascending part of the Gaussian function from v_{L1} ($=\mu_{\lambda 1r}$) to v_{L2} ($=\mu_{\lambda 2r}$) when $t < t_2$, and a constant v_{L2} that satisfies $v_{L2} = \mu_{\lambda 3r} = \mu_{\lambda 2r}$ when $t \geq t_2$. Specifically, $\lambda_{Ar}(t)$ is described as

$$\lambda_{Ar}(t) = \begin{cases} v_{L1} + (v_{L2} - v_{L1}) \exp(v_{\exp}), & t < t_2 \\ v_{L2}, & t \geq t_2 \end{cases} \quad (30)$$

where $v_{L2} > v_{L1}$, $v_{\exp} = -(t - t_2)^2 / (2v_{L3}^2)$, and v_{L1} , v_{L2} , and v_{L3} are positive constants. Its trajectory is shown in Fig. 11 of Appendix. Based on (30) and Fig. 11 of Appendix, it is clear that this example (30) satisfies Conditions (CB2) and (CB3).

Proof. : By taking the derivative of S^* in (26) and combining (11) and (28), the r th term of $\dot{S}^*$ satisfies

$$\dot{S}_r^* = c_{2r}\tilde{\sigma}_{1r} + \tilde{\sigma}_{2r} + c_{3r}\dot{\tilde{\sigma}}_{1r} + \tilde{\sigma}_{1r} - \lambda_{Ar}(t)S_r^* - \lambda_{Br}\text{sat}(S_r^*) + \alpha_0 S_{0r}(t) \quad (31)$$

$$S_{0r}(t) := S_r(0)\exp(-\alpha_0 t).$$

Substituting (8), (10) and (11) into (31) gives

$$\dot{S}_r^* = (c_{2r} + c_{3r}l_{1r}\dot{h}_i(t) + l_{12r}\dot{h}_i^2(t))\tilde{\sigma}_{1r} + \tilde{\sigma}_{2r} + c_{3r}\dot{\tilde{\sigma}}_{1r} + \tilde{\sigma}_{1r} - \lambda_{Ar}(t)S_r^* - \lambda_{Br}\text{sat}(S_r^*) + \alpha_0 S_{0r}(t) \quad (32)$$

Here, we define the Lyapunov function

$$V_S(S_r^*) = S_r^{*2} / 2 \quad (33)$$

We know that $\dot{V}_S(S_r^*) = \dot{S}_r^* S_r^*$. Based on (22), multiplying $\dot{S}_r^*$ by S_r^* produces

$$\begin{aligned} S_r^* \dot{S}_r^* &\leq |S_r^*|[(c_{2r} + c_{3r}l_{1r}\dot{h}_i(t) + l_{12r}\dot{h}_i^2(t) + c_{3r} + 2)\|z_i\| + \mu_1(c_{3r} + 1)|S_r^*| \\ &\quad - \lambda_{Br}|\text{sat}(S_r^*)| \cdot |S_r^*| - \lambda_{Ar}(t)|S_r^*|^2 + \alpha_0 S_r(0)|S_r^*|] \\ &\leq -|S_r^*|[-(c_{2r} + c_{3r}l_{1r}\dot{h}_i(t) + l_{12r}\dot{h}_i^2(t) + c_{3r} + 2)\|z_i\| \\ &\quad + \lambda_{Ar}(t)|S_r^*| + \lambda_{Br}|\text{sat}(S_r^*)| - (c_{3r} + 1)\mu_1 - \alpha_0 S_r(0)] \\ &\leq -\lambda_{Ar}(t)|S_r^*|(|S_r^*| - \chi_{B0}) \\ \chi_{B0} &= [(c_{2r} + c_{3r}l_{1r}\dot{h}_i(t) + l_{12r}\dot{h}_i^2(t) + c_{3r} + 2)\|z_i\| \\ &\quad + \alpha_0 S_r(0) + (c_{3r} + 1)\mu_1 - \lambda_{Br}|\text{sat}(S_r^*)|] / \lambda_{Ar}(t) \end{aligned} \quad (34)$$

Thus, after a sufficiently long time, we obtain that $\dot{V}_S(S_r^*) \leq 0$ and $|S_r^*| \leq \chi_{B0}$. From (22), when $t \geq T_A$, z_i is ultimately bounded and $\|z_i\| \leq \chi_{A1}$. According to (22) and (34), we find that S_r^* converges only after $t \geq T_A$. Hence, after a sufficiently long time that $t \geq T_B$ ($T_A \leq T_B < \infty$),

$|S_r^*|$ satisfies

$$\begin{aligned} |S_r^*| &\leq \chi_{B1} \\ \chi_{B1} &= \chi_{B0} \|z_i\| = \chi_{A1} = [(c_{3r} + 1)\mu_1 - \lambda_{Br}|\text{sat}(S_r^*)| + \alpha_0 S_r(0) \\ &\quad + (c_{2r} + c_{3r}l_{1r}\dot{h}_i(t) + l_{12r}\dot{h}_i^2(t) + c_{3r} + 2)\chi_{A1}] / \lambda_{Ar}(t) \end{aligned} \quad (35)$$

T_B is the unknown convergent time of the improved sliding-mode surface S_r^* . This means that S_r^* is ultimately bounded.

Since t_1 and t_2 are at the initial period of system response, when the error system (3) runs for a sufficiently long time, we obtain that $\dot{h}_i(t) = \mu_3$ and $\lambda_{Ar}(t) = \mu_{\lambda 3r}$. Based on (33) and (35), the boundary of $|S_r^*|$ specifically narrows to

$$\begin{aligned} |S_r^*| &\leq \chi_{B2} \\ \chi_{B2} &= [(c_{2r} + c_{3r}l_{1r}\mu_3 + l_{12r}\mu_3^2 + c_{3r} + 2)\chi_{A2} + (c_{3r} + 1)\mu_1 \\ &\quad + \alpha_0 S_r(0) - \lambda_{Br}|\text{sat}(S_r^*)|] / \mu_{\lambda 3r}. \end{aligned} \quad (36)$$

Next, based on (3), (11) and (25), the r th term S_r^* of S^* is rewritten as

$$S_r^* = c_{1r}e_{1r} + c_{2r}\dot{e}_{1r} + c_{3r}\ddot{e}_{1r} + e_{1r}^{(3)} - c_{3r}\tilde{\sigma}_{1r} - \tilde{\sigma}_{1r} - S_{0r}(t) \quad (37)$$

We rewrite (37) to be the following differential equation:

$$\begin{aligned} e_{1r}^{(3)} + c_{3r}\ddot{e}_{1r} + c_{2r}\dot{e}_{1r} + c_{1r}e_{1r} &= \varpi_{0r}(t) \\ \varpi_{0r}(t) &= c_{3r}\tilde{\sigma}_{1r} + \tilde{\sigma}_{1r} + S_{0r}(t) + S_r^* \end{aligned} \quad (38)$$

The solution of (38), e_{1r} , is composed of general solution and special solution two terms, which are defined as e_{G1r} and e_{P1r} respectively. To analyze the boundedness of e_{1r} , we first discuss e_{G1r} and e_{P1r} one by one.

Let $E_r := [e_{1r}, \dot{e}_{1r}, \ddot{e}_{1r}]^T$. Based on (27) and (38), we obtain that

$$\dot{E}_r = L_{Cr}E_r + D_r, \quad D_r = \varpi_{0r}(t)[0, 0, 1]^T, \quad r = 1, 2, \dots, n \quad (39)$$

Since (38) belongs to a nonhomogeneous linear differential equation with constant coefficients, based on (39), we obtain that $\Lambda_{vr}(L_{Cr})$ is the eigenvalue of the homogeneous equation of (38). And based on Condition (CB1), we know that, for the homogeneous equation of (38), the real part of each eigenvalue, which is defined as α_{Avr} ($v = 1, 2, 3$, $r = 1, 2, \dots, n$), is negative.

By solving the differential Eq. (38), the fundamental system of solutions for (38) has the following four situations: (G1) $\exp(\alpha_{A1r}t)$, $\exp(\alpha_{A2r}t)$, and $\exp(\alpha_{A3r}t)$, if each $\Lambda_{vr}(L_{Cr}) = \alpha_{Avr}$ and all the eigenvalues are not equal. (G2) $\exp(\alpha_{A1r}t)$, $t \exp(\alpha_{A1r}t)$, and $\exp(\alpha_{A3r}t)$, if each $\Lambda_{vr}(L_{Cr}) = \alpha_{Avr}$ and there exist two equal eigenvalues in $\Lambda_{vr}(L_{Cr})$, which are set as $\Lambda_{1r}(L_{Cr})$ and $\Lambda_{2r}(L_{Cr})$. (G3) $\exp(\alpha_{A1r}t)$, $t \exp(\alpha_{A1r}t)$, and $t^2 \exp(\alpha_{A1r}t)$, if each $\Lambda_{vr}(L_{Cr}) = \alpha_{Avr}$ and all the eigenvalues are equal. (G4) $\exp(\alpha_{A1r}t)$, $\sin \beta_{A2r} \exp(\alpha_{A2r}t)$, and $\cos \beta_{A2r} \exp(\alpha_{A2r}t)$, if there exists one eigenvalue, which is set as $\Lambda_{1r}(L_{Cr})$, that satisfies $\Lambda_{1r}(L_{Cr}) = \alpha_{A1r}$ and the other two eigenvalues $\Lambda_{2r}(L_{Cr})$ and $\Lambda_{3r}(L_{Cr})$ are conjugated with imaginary parts $\pm \beta_{A2r}$.

Since α_{Avr} is negative, from (G1) - (G4), it is clear that $\lim_{t \rightarrow \infty} \exp(\alpha_{Avr}t) = 0$, and $\lim_{t \rightarrow \infty} t^i \exp(\alpha_{A1r}t) = 0$. Hence, the general solution of (38), e_{G1r} , satisfies

$$\lim_{t \rightarrow \infty} e_{G1r} = 0. \quad (40)$$

Then, combining (22) and (35), after a sufficiently long time that $t \geq T_B$, the boundary of $|\varpi_{0r}(t)|$ narrows to

$$\begin{aligned} |\varpi_{0r}(t)| &\leq |c_{3r}\tilde{\sigma}_{1r} + \tilde{\sigma}_{1r}| + S_{0r}(t) + |S_r^*| \\ &\leq (c_{3r} + 1)\chi_{A1} + S_{0r}(t) + \chi_{B1} \end{aligned} \quad (41)$$

Since $\dot{h}_i(t) - 12\Lambda_{\max}(P_i)\dot{h}_i(t)/h_i(t) \geq 1$, $1 \leq \dot{h}_i(t) \leq \mu_2$ and $\lambda_{Ar}(t) \geq \mu_{\lambda 1r}$, based on (22) and (35), (41) becomes

$$|\varpi_{0r}(t)| \leq \varpi_{1r} + S_{0r}(t) \quad (42)$$

where

$$\begin{aligned} \varpi_{1r} &= (c_{3r} + 1)\chi_{A3} + \chi_{B3}, \quad \chi_{A3} = 2\mu_1\Lambda_{\max}(P_i) \\ \chi_{B3} &= [(c_{2r} + c_{3r}l_{1r}\mu_2 + l_{12r}\mu_2^2 + c_{3r} + 2)\chi_{A3}\alpha_0 S_r(0) + (c_{3r} + 1)\mu_1] / \mu_{\lambda 1r} \end{aligned} \quad (43)$$

Based on (31), (38), (42), and Condition (CB1) with respect to α_0 , and combined with the mathematical knowledge of solving differential Eq. (38), $|e_{p1r}|$ is bounded by

$$|e_{p1r}| \leq \chi_{C1} \cdot \chi_{C1} = \frac{\varpi_{1r}}{c_{1r}} + \frac{S_{0r}(t)}{c_{1r} - c_{2r}\alpha_0 + c_{3r}\alpha_0^2 - \alpha_0^3} \quad (44)$$

$$\lim_{t \rightarrow \infty} \frac{S_{0r}(t)}{c_{1r} - c_{2r}\alpha_0 + c_{3r}\alpha_0^2 - \alpha_0^3} = 0.$$

Then, we apply the discussion results in (40) and (44) to e_{1r} . Based on (42)–(44), we find that e_{1r} converges only after $t \geq T_B \geq T_A$. Therefore, after a sufficiently long time $t \geq T_C$ ($T_A \leq T_B \leq T_C < \infty$),

$$|e_{1r}| \leq |e_{G1r}| + \chi_{C1} \quad (45)$$

T_C is the unknown convergent time of error signal e_{1r} . This means that e_1 is ultimately bounded.

Since t_1 and t_2 are at the initial period of system response, when the error system (3) runs for a sufficiently long time, $\hat{h}_i(t) = \mu_3$ and $\lambda_{Ar}(t) = \mu_{\lambda 3r}$. So, we further derive the specific ultimate boundary value of $|e_{1r}|$.

Substituting (23) and (36) into (41) gives

$$|\varpi_{0r}(t)| \leq \varpi_{3r} + S_{0r}(t), \quad \varpi_{3r} = (c_{3r} + 1)\chi_{A2} + \chi_{B2} \quad (46)$$

Based on (38), (44), (46), and Condition (CB1) with respect to α_0 , we find $|e_{p1r}|$ is specifically bounded by

$$|e_{p1r}| \leq \chi_{C2}, \quad \chi_{C2} = \frac{\varpi_{3r}}{c_{1r}} + \frac{S_{0r}(t)}{c_{1r} - c_{2r}\alpha_0 + c_{3r}\alpha_0^2 - \alpha_0^3} \quad (47)$$

Hence, following (40) and (47), we obtain that the boundary of $|e_{1r}|$ specifically narrows to

$$|e_{1r}| \leq |e_{G1r}| + \chi_{C2}, \quad \lim_{t \rightarrow \infty} |e_{1r}| \leq \varpi_{3r}/c_{1r} \quad (48)$$

Therefore, the ultimate boundedness of e_1 is guaranteed. ■

Remark 7. For the whole design parameters in the proposed method (i.e., l_{ivr} , c_{vr} , λ_{Br} , α_0 , ε , $\hat{h}_i(t)$, $\lambda_{Ar}(t)$, and the extra design constants about $\hat{h}_i(t)$ and $\lambda_{Ar}(t)$), they need to be chosen under the design principles of the following two aspects. On the one hand, as mentioned in Sections 3 and 4, all of the design parameters need to satisfy their definitions, Conditions (CA1) - (CA2), Conditions (CB1) - (CB3), Remark 3, and Remark 6. On the other hand, we give priority to improving tracking accuracy by setting constant design parameters under the following design principles to narrow the boundary of $|e_{1r}|$. Based on (31) and (47), the exponential term of (47) can be reduced quickly by setting α_0 to be a not small enough value. To reduce ϖ_{3r}/c_{1r} in (48), based on (23), (36), and (46)–(48), $\mu_{\lambda 3r}$ and c_{1r} need to be set as larger ones, while c_{3r} and $\Lambda_{\max}(P_i)$, which are affected by l_{ivr} , are the opposites. When the error state $e_{kr}(0) \neq 0$, the mismatched initial state exacerbates the peaking phenomenon, and then $\mu_{\lambda 2r}$ ($< \mu_{\lambda 3r}$) is recommended to be less than 1. From (29), a smaller ε can reduce chattering. These above design principles give the general path of design parameter selection. According to these design principles, the optimized values of all the design parameters are further obtained based on the experience of multiple tests.

Remark 8. Compared with traditional SMC, the ISMC has the following novelties:

- S^* includes $\hat{\sigma}_1$ and $\hat{\sigma}_1$, which is helpful to design control law (28) to overcome mismatched disturbances vector σ_1 and improve tracking accuracy.

- S^* includes the vector $-S_0(t)$ in (25). Based on (25) and (28), including $-S_0(t)$ in (25) is useful to eliminate the effect on the initial values of $\hat{\sigma}_1$, $\hat{\sigma}_1$, and S^* in (25) and to alleviate peaking phenomenon of torque at the initial period.

- The ISMC law (28) includes the time-varying design matrix $\lambda_A(t)$, which is one of the effective ways to balance the contradiction between high tracking accuracy and suitable torque under suitable $\lambda_A(t)$.

5. Simulation and experiment

In this section, numerical examples and an experiment are provided for the UFJM using the TVDO-based ISMC method. To illustrate the effects of control law in improving tracking control performance, we consider a special case of the proposed method by choosing constant design parameters instead of time-varying design parameters. To illustrate the superiority of the proposed method, in the simulation and experiment, we try our best to compare it with other methods under the same physical environments. Specifically, in the simulation, we employ the same physical parameters, initial states, and target trajectories as those in the comparative methods, and we set the same or larger uncertainties from the comparative methods.

5.1. Example 1: uncertain single-link FJM

In this example, we present the numerical results of the TVDO-based ISMC method for a vertical UFJM with single-link. In the dynamic model of the UFJM with single-link, $M(q) = J_1$, $C(q, \dot{q})\dot{q} = 0$, $G(q) = m_1 g L_{c1} \sin q_1$, $I_A = I_{A1}$, $K = K_1$, and $B_A = B_{A1}$, where L_{c1} is the length from the link's center of mass to the flexible joint, and $g = 9.8 \text{ N/kg}$ is the gravity constant.

We first compare the proposed method with the DO-based SMC method (Ginoya et al., 2014) to illustrate the effects of control law in improving tracking control performance. According to the uncertainties range mentioned in Ginoya et al. (2014), we use its maximized uncertainties $d_1 = 2[-4.5/J_1 - K_1/J_1 \cdot (q_1 - \theta_1)]$ and $d_2 = 2[K_1/I_{A1} \cdot (q_1 - \theta_1) - B_{A1}/I_{A1} + 1/I_{A1} \cdot (\tau_1 + \sin(2t) + 1)]$.

The design parameters of the proposed method are set as $l_{i1r} = 300$, $l_{i2r} = 90$, $l_{i3r} = 16$, $c_{1r} = 125$, $c_{2r} = 75$, $c_{3r} = 15$, $\lambda_{Br} = 0.5$, $\alpha_0 = 1$, and $\varepsilon = 0.1$. For this set of simulation, $\hat{h}_i(t)$ is set as $\hat{h}_i^{\#}(t)$ and $\lambda_{Ar}(t)$ is set as $\lambda_{Ar}^{\#}(t)$, where $t_1 = t_2 = 2 \text{ s}$, $v_{H1} = 5$, $v_{H2} = 1$, $v_{L1} = 1$, $v_{L2} = 40$, and $v_{L3} = 0.2$.

Since the Ginoya et al. (2014) mainly studied the vibration suppression problem, its target trajectory is actually chosen as $q_{d1} = 0 \text{ rad}$. From the numerical results in Figures 9 and 10 of Ginoya et al. (2014), its maximum tracking error in the steady state is about 0.002 rad. We present the numerical results of our proposed method in Fig. 2(b) - (d). It is clear that the proposed method has higher tracking accuracy for the uncertain single-link FJM and stronger vibration suppression capability under the effect of model uncertainties and disturbance uncertainties.

Next, we compare our proposed method with the equivalent-input-disturbance- (EID-) based method (Yan et al., 2021) and analyze the effects of our control law. Here, we set the design parameters of the TVDO-ISMC method to be $l_{i1r} = 100$, $l_{i2r} = 40$, $l_{i3r} = 5$, $c_{1r} = 40$, $c_{2r} = 20$, $c_{3r} = 5$, $\lambda_{Br} = 0.5$, $\alpha_0 = 1$ and $\varepsilon = 0.1$. $\hat{h}_i(t)$ is chosen to satisfy (24), in which $t_1 = 5 \text{ s}$, $v_{H1} = 300$, and $v_{H2} = 30$. $\lambda_{Ar}(t)$ is chosen to satisfy (30), in which $t_2 = 1 \text{ s}$, $v_{L1} = 0.1$, $v_{L2} = 80$, and $v_{L3} = 0.2$. To further analyze the effects of our control law when tracking a time-varying trajectory, we consider a special case by choosing constant design parameters instead of time-varying design parameters for the proposed method, i.e., setting $\hat{h}_i(t) = 30$, $\lambda_{Ar}(t) = 80$ and $S_0(t) = 0$.

To better describe the tracking accuracy, the following performance index is given to be

$$\Xi_{\max} := \max(|e_{kr}|) / [\max(q_{dr}) - \min(q_{dr})] \times 100\% \quad (49)$$

which denotes the percentage of maximum error. From (49), the smaller $\Xi_{\max}$ in the steady state means the higher tracking accuracy of the UFJM.

Based on the numerical results in Fig. 2 of this paper and Fig. 5 of Yan et al. (2021), $\Xi_{\max}$ in the steady state are 0.01% (Proposed-1), 0.01% (Constant-1), and 3.00% (EID). However, the torque of the “Constant-1” case has a larger peak value, which is unrealistic in practice. Hence, our proposed method has better tracking control performance under suitable torque than the EID-based method. Meanwhile, it is necessary to consider time-varying $\hat{h}_i(t)$, $\lambda_{Ar}(t)$, and $S_0(t)$ in our control law.

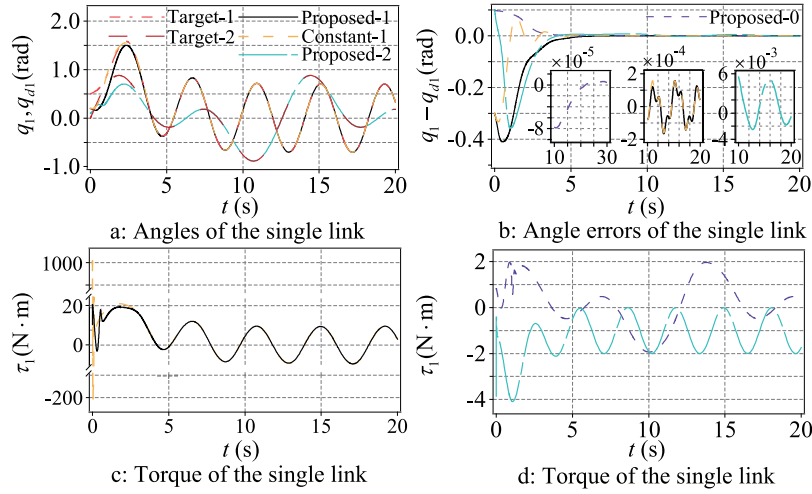


Fig. 2. Tracking results for uncertain single-link FJM (“Proposed-0”, “Proposed-1” and “Proposed-2” employ the proposed method and belong to the cases compared with Ginoya et al. (2014), Yan et al. (2021) and Hu et al. (2021), respectively; “Target-1” and “Target-2” denote the target trajectories in Yan et al. (2021) and Hu et al. (2021); “Constant-1” denotes the special case of the “Proposed-1” using constant design parameters).

Then, we compare the TVDO-based ISMC method with the nonlinear disturbance-observer- (NDO-) based dynamic surface control (DSC) method with IT2NNA (Hu et al., 2021). The design parameters of the proposed method are set to be $l_{11r} = 80$, $l_{12r} = 32$, $l_{13r} = 4$, $c_{1r} = 40$, $c_{2r} = 20$, $c_{3r} = 5$, $\lambda_{Br} = 0.5$, $\alpha_0 = 1$, and $\varepsilon = 0.1$. $\hat{h}_i(t)$, and $\lambda_{Ar}(t)$ are chosen to satisfy (24) and (30), respectively, where $t_1 = 5$ s, $v_{H1} = 2$, $v_{H2} = 1$, $t_2 = 1$ s, $v_{L1} = 0.02$, $v_{L2} = 200$, and $v_{L3} = 0.02$. We compare the numerical results of the proposed method shown in Fig. 2 with those of Hu et al. (2021) (see Figures 6 and 8 of Hu et al., 2021). Clearly, the torque of the proposed method is less than 2 N.m and is in a suitable range. It can be calculated that, in the steady state, $\Xi_{\max}$ of the proposed method is 0.33%, and $\Xi_{\max}$ of the NDO-based DSC method with IT2NNA is more than 0.91%. Thus, our proposed method has better control performance than the NDO-based DSC method with IT2NNA.

5.2. Example 2: uncertain two-link FJM

In this example, we present the numerical results of the proposed method for a UFJM with two links. In the dynamic model of the UFJM with two links, $K = \text{diag}(K_1, K_2)$, $I_A = \text{diag}(I_{A1}, I_{A2})$, $B_A = \text{diag}(B_{A1}, B_{A2})$, $G(q) = [-(m_1 + m_2)L_1 g \sin q_1, -m_2 L_2 g \sin q_2]^T$, $M(q) = [(m_1 + m_2)L_1^2, m_1 L_1 L_2 \cos(q_1 - q_2); m_1 L_1 L_2 \cos(q_1 - q_2), m_2 L_2^2]$, and $C(q, \dot{q})\dot{q} = -m_2 L_1 L_2 \sin(q_2 - q_1)[\dot{q}_2^2, \dot{q}_1^2]^T$.

We contrast our proposed method with the adaptive fuzzy command filtered (AFCF) method (Ling et al., 2020). In the simulation, the AFCF method did not consider uncertainties for the two-link FJM. Here, we consider the effect of model uncertainties and disturbance uncertainties. We set the uncertainties as $d_1 = -0.2[C(q, \dot{q})\dot{q} + G(q) + K \cdot (q - \theta)] - 0.2[\sin(10t + 1), \sin(15t)]^T$ and $d_2 = -0.1[-K \cdot (q - \theta) + B_A \dot{\theta}] - 0.3[\cos(8t), \sin(10t)]^T$.

The design parameters of the TVDO-based ISMC method are chosen to be $l_{11r} = 150$, $l_{12r} = 60$, $l_{13r} = 6$, $c_{1r} = 60$, $c_{2r} = 30$, $c_{3r} = 5$, $\lambda_{Br} = 0.5$, $\alpha_0 = 1$, and $\varepsilon = 0.1$. We choose the time-varying design parameters $\hat{h}_i(t)$ and $\lambda_{Ar}(t)$ to satisfy (24) and (30), respectively, where $t_1 = 4$ s, $v_{H1} = 10$, $v_{H2} = 2$, $t_2 = 1$ s, $v_{L1} = 0.2$, $v_{L2} = 200$, and $v_{L3} = 0.2$. To further analyze the effects of our control law when tracking time-varying trajectories of the UFJM with two links, we consider the special case by choosing constant design parameters for the proposed method, i.e., setting $\hat{h}_i(t) = 2$, $\lambda_{Ar}(t) = 200$ and $S_0(t) = 0$.

The simulation results of the proposed method and the special case are presented in Fig. 3, and the simulation results of the AFCF method can be seen in Figs. 6–8 of Ling et al. (2020). It can be calculated that $\Xi_{\max}$ in the steady state for two links are 0.01% (first link,

Proposed), 0.01% (second link, Proposed), 8.75% (first link, AFCF), and 7.90% (second link, AFCF). Clearly, the tracking accuracy of the proposed method is higher than that of the AFCF method. As is shown in Fig. 3(f), the torque of the special case “Constant” exists large peaking phenomenon. Thus, our proposed method has better tracking control performance within suitable torque than the AFCF method. It is important to consider time-varying $\hat{h}_i(t)$, $\lambda_{Ar}(t)$, and $S_0(t)$ in our control law to get good tracking control performance.

To further demonstrate the superiority of the TVDO-based ISMC method, we compare it with a finite-time DO-based feedback control (FTDOBFC) method in Wang, Zhang, et al. (2021). For the proposed method, we choose its design parameters as $l_{11r} = 200$, $l_{12r} = 80$, $l_{13r} = 10$, $c_{11} = 4c_{12}/3 = 20$, $c_{21} = 4c_{22}/3 = 10$, $c_{31} = 4c_{32}/3 = 2.5$, $\lambda_{Br} = 0.5$, $\alpha_0 = 1$, and $\varepsilon = 0.1$. The design parameters of $\hat{h}_i(t)$ and $\lambda_{Ar}(t)$ under (24) and (30) are set to be $t_1 = 5$ s, $v_{H1} = 10$, $v_{H2} = 3$, $t_2 = 3$ s, $v_{L1} = 0.1$, $v_{L3} = 0.2$, and v_{L2} in $\lambda_{A1}(t)$ and $\lambda_{A2}(t)$ equals 10 and 6, respectively.

The simulation results of the proposed method and the FTDOBFC method are presented in Fig. 4 of this paper and Fig. 3 of Wang, Zhang, et al. (2021), respectively. Through calculation, $\Xi_{\max}$ in the steady state are 0.21% (first link, Proposed), 0.43% (second link, Proposed), 2.08% (first link, FTDOBFC), and 5% (second link, FTDOBFC). Meanwhile, the torques of the FTDOBFC method have large initial peak values, which do not occur with the proposed method. Hence, the TVDO-based ISMC method shows better tracking control performance in the transient state and steady state than the FTDOBFC method.

5.3. Example 3: uncertain three-link FJM

To show the universality of the proposed method and further illustrate the effect of time-varying design parameters in improving tracking control performance of the UFJM, we set a series of cases for a UFJM with three links.

In the dynamic model of the UFJM with three links, $M(q) = [a_1, a_2 \tilde{c}_{12}, a_3 \tilde{c}_{13}; a_2 \tilde{c}_{12}, a_4, a_5 \tilde{c}_{23}; a_3 \tilde{c}_{13}, a_5 \tilde{c}_{23}, a_6]$, $C(q, \dot{q})\dot{q} = [a_2 \tilde{s}_{12} \dot{q}_2^2 + a_3 \tilde{s}_{13} \dot{q}_3^2, -a_2 \tilde{s}_{12} \dot{q}_1^2 + a_5 \tilde{s}_{23} \dot{q}_3^2, -a_3 \tilde{s}_{13} \dot{q}_1^2 - a_5 \tilde{s}_{23} \dot{q}_2^2]^T$, $G(q) = [-a_7 \sin q_1, a_8 \sin q_2, a_9 \sin q_3]^T$, $K = \text{diag}(K_1, K_2, K_3)$, $I_A = \text{diag}(I_{A1}, I_{A2}, I_{A3})$, and $B_A = \text{diag}(B_{A1}, B_{A2}, B_{A3})$, in which $a_1 = J_{c1} + m_1 L_{c1}^2 + L_1^2(m_2 + m_3)$, $a_2 = L_1(m_2 L_{c2} + m_3 L_{c3})$, $a_3 = m_3 L_1 L_{c3}$, $a_4 = J_{c2} + m_2 L_{c2}^2 + m_3 L_{c3}^2$, $a_5 = m_3 L_2 L_{c3}$, $a_6 = J_{c3} + m_3 L_{c3}^2$, $a_7 = [L_{c1} m_1 + L_1(m_2 + m_3)]g$, $a_8 = (m_2 L_{c2} + m_3 L_{c3})g$, $a_9 = m_3 L_{c3}g$, $\tilde{s}_{ib} = \sin(q_i - q_b)$, $\tilde{c}_{ib} = \cos(q_i - q_b)$, and L_{ci} is the length from the i -th link's center of mass to its flexible joint, actuator inertia, and link inertia about its center of mass ($i = 1, 2, 3$; $b = 1, 2, 3$).

Here, we choose the physical parameters of an uncertain three-link FJM as $m_1 = 2.4$ kg, $m_2 = 2.1$ kg, $m_3 = 2.3$ kg, $L_1 = 0.9$ m,

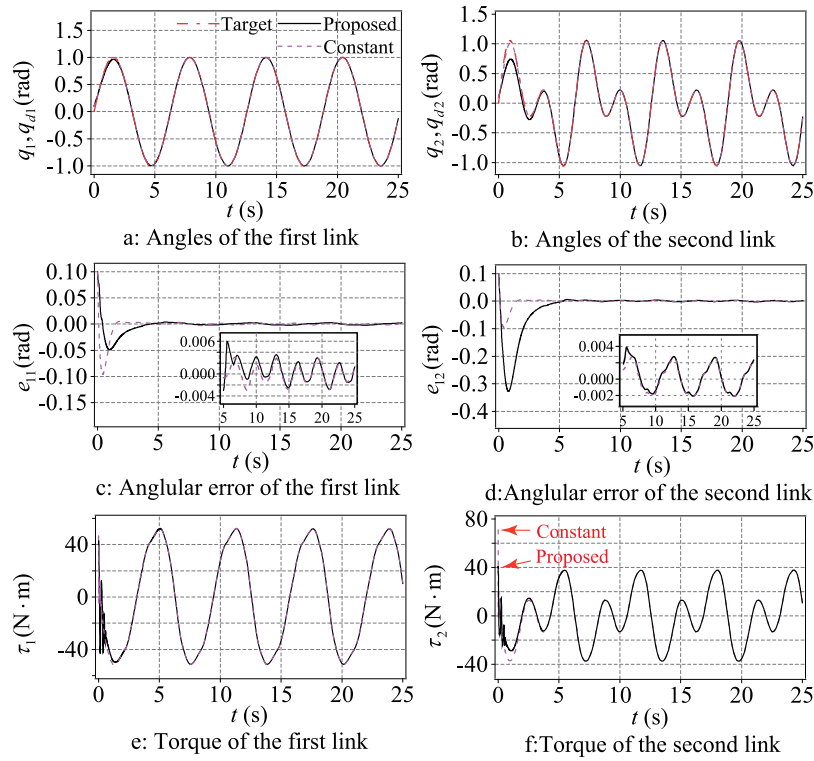


Fig. 3. Tracking results for uncertain two-link FJM (“Proposed” employs the proposed method and belongs to the case compared with Ling et al. (2020); “Target” denotes the target trajectories in Ling et al. (2020); “Constant” denotes the special case of the “Proposed” using constant design parameters).

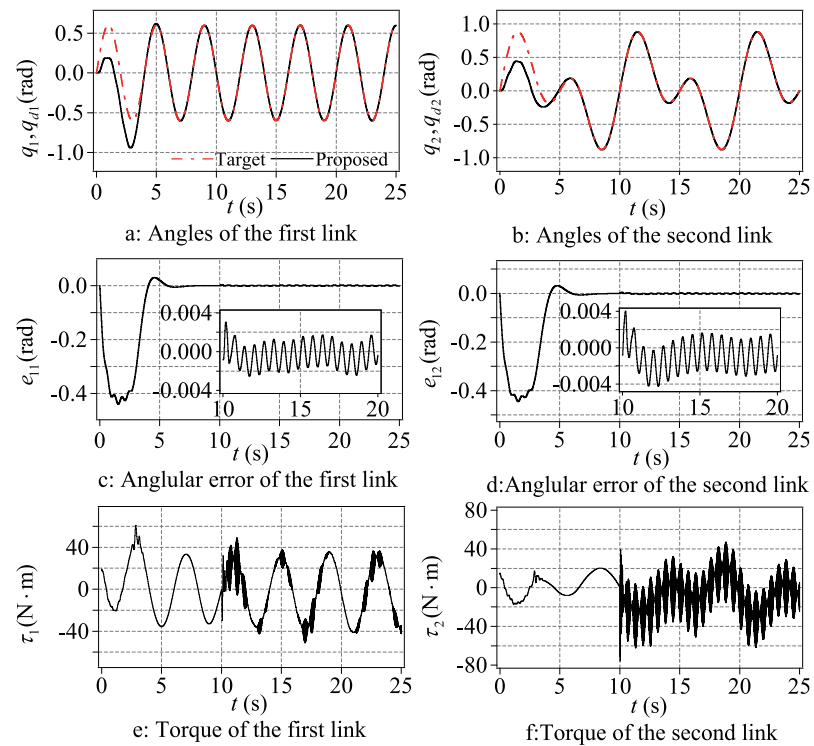


Fig. 4. Tracking results for uncertain two-link FJM (“Proposed” employs the proposed method and belongs to the case compared with Wang, Zhang, et al., 2021; “Target” denotes the target trajectories in Wang, Zhang, et al., 2021).

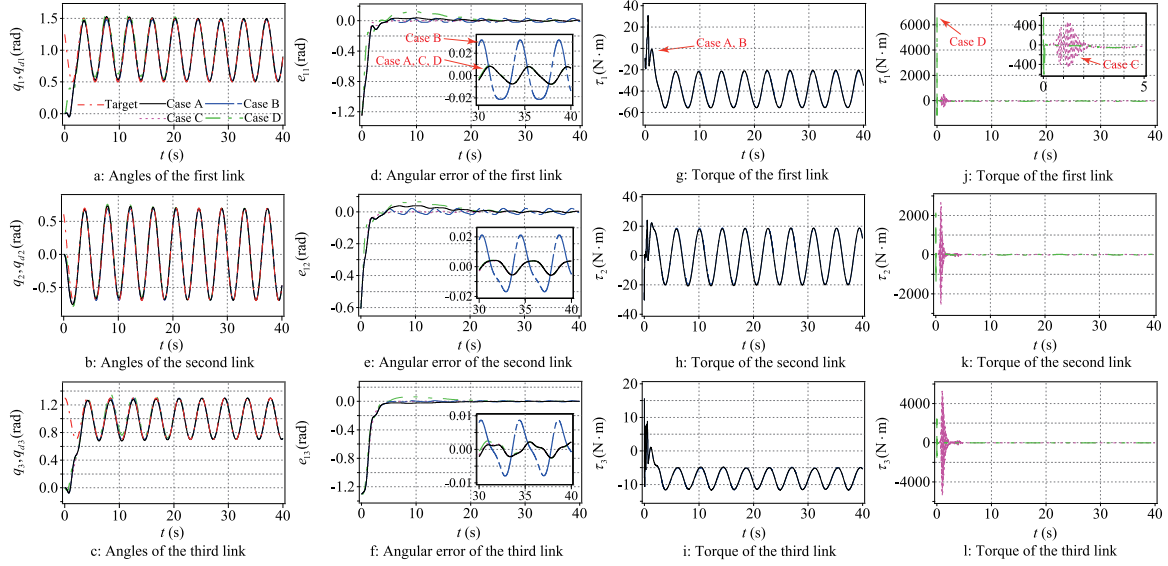


Fig. 5. Tracking control results for UFJM with three links by the TVDO-based ISMC method under Cases A-D.

Table 1
Cases A-E for the UFJM with three links.

Cases	A (Proposed)	B	C	D	E (Cancel)
$\hat{h}_i(t)$	$\hat{h}_i^*(t)$	100	1	$\hat{h}_i^*(t)$	$\hat{h}_i^*(t)$
$\lambda_{Ar}(t)$	$\lambda_{Ar}^*(t)$	$\lambda_{Ar}^*(t)$	$\lambda_{Ar}^*(t)$	80	0.1

$L_2 = 0.8$ m, $L_3 = 0.9$ m, $L_{1r} = 0.45$, $L_{2r} = 0.4$, $L_{3r} = 0.45$, $J_1 = 0.162$ kg m², $J_2 = 0.112$ kg m², $J_3 = 0.155$ kg m², $K_r = 10$ N/m/rad, $I_{Ar} = 0.7$ kg m², and $B_{Ar} = 0.02$. The initial states are set as $q_r(0) = 0$ rad, and $\theta_r(0) = 0$ rad. The uncertainties are set as $d_1 = [0.1 \sin(q_1 - q_3)\theta_3^2 + 0.1 \cos(q_1) + 0.1 \sin(q_1 - q_2)\theta_2^2, 0.1 \cos(4t) + 0.1 \sin(q_3)]^T$, $d_2 = [0.1 \cos(3t) + 0.1(q_1 - \theta_1), 0.1\dot{\theta}_2 + 0.1 \cos(5t), 0.1 \cos(2t) + 0.1\dot{\theta}_3]^T$. We make link angles track the target angles that $q_{d1} = 0.5 \cos(1.5t + \pi/3) + 1$, $q_{d2} = 0.7 \sin(1.5t + 2\pi/3)$, $q_{d3} = 0.3 \cos(1.5t) + 1$.

We use Cases A-C and Cases A, D, and E in Table 1 to illustrate the important roles of using the time-varying $\hat{h}_i(t)$ and $\lambda_{Ar}(t)$, respectively. Among the Cases, the time-varying $\hat{h}_i(t)$ and $\lambda_{Ar}(t)$ are set as $\hat{h}_i^*(t)$ and $\lambda_{Ar}^*(t)$, in which $t_1 = 5$ s, $t_2 = 1$ s, $v_{H1} = 100$, $v_{H2} = 1$, $v_{L1} = 0.1$, $v_{L2} = 80$, and $v_{L3} = 0.2$. Since $\lambda_{Ar}(t)$ in Case E is too small to allow the system to reach a steady state, we do not present its simulation results in Fig. 5. As shown in Fig. 5, all Cases A-D can make the links track corresponding target trajectories. However, comparing Cases B and C with Case A, Case B has worse tracking accuracy, and Case C has larger peak values and violent shaking in the torques. Comparing Cases D and E with Case A, Case D has larger torques range. The percentage of maximum errors in steady state $\varepsilon_{\max}$ of Case A for three links are 0.77% (first link), 0.57% (second link), and 0.43% (third link). By contrast and calculation, Case A that uses time-varying $\hat{h}_i(t)$ and $\lambda_{Ar}(t)$ can coordinate the contradiction between high tracking accuracy and suitable torques obviously.

5.4. Experiment: uncertain single-link FJM

Here, we take an uncertain single-link FJM as the practical control object and conduct an experiment to illustrate the effectiveness of the TVDO-based ISMC method. As shown in Fig. 6, the link with load is connected with a flexible joint in the form of a torsional spring,

and the flexible joint is directly actuated by a Panasonic MINAS A6 series AC servo motor MHMF042L1U2M. The servo motor works in conjunction with its dedicated motor driver, Panasonic servo motor driver MBDLT25SF. q_1 is measured by a photoelectric encoder, and θ_1 is measured by a motor encoder. The signal transmission is achieved by an I/O module PCIe-6361 made by National Instruments Corporation. The implementation of this experiment is in a PC (Intel Core i7-12700K; 32G) and based on MATLAB/Simulink software. The implementation procedures of the proposed method for the UFJM are listed as follows.

Step 1: Obtain the physical parameters of this UFJM through measurement and calculation. The physical parameters are $m_1 = 0.48$ kg, $L_{c1} = 0.31$ m, $J_1 = 0.039$ kg m², $K_1 = 0.1$ N/m/rad, $I_{A1} = 56$ E-6 kg m², and $B_{A1} = 0.001$.

Step 2: Using MATLAB/Simulink software, we write the program of the control method.

Step 3: Set the sampling period as 0.005 s, make q_1 , θ_1 , $\dot{q}_1$ and $\dot{\theta}_1$ start from zero, and make q_1 track the target trajectory that

$$q_{1d} = \begin{cases} 0.25 \sin(0.8t - 0.5\pi) + 0.25, & t \leq 1.25 \text{ s} \\ 0.05 \sin(0.2t - 0.75\pi) + 0.5, & t > 1.25 \text{ s} \end{cases} \quad (50)$$

Step 4: We transfer the angle measurement signals q_1 and θ_1 from the encoders to the PC via an I/O module, use the toolbox function “Encoder Input” in Simulink to obtain the measurement signals in real time, and then differentiate them in Simulink to attain $\dot{q}_1$ and $\dot{\theta}_1$, respectively.

Step 5: According to the measured angles and velocities, we use the control law (28) to calculate τ_1 . Then, we use the toolbox function “Analog Output” in Simulink and the I/O module to transfer the control torque τ_1 to the motor driver. Meanwhile, we make the motor driver work in its built-in Torque Control Mode and configure it to make the servo motor generate the actual torque that is approximately equal to τ_1 .

As shown in Fig. 7, the motor driver receives an analog voltage signal at a certain proportion of τ_1 from the I/O module and then employs it to make the servo motor generate the actual actuator torque $\tau_{1|actual}$ under the torque control mode. Since the Torque Control Mode of the motor driver includes a control algorithm that takes motor dynamics into account, this motor driver has the ability to make $\tau_{1|actual} \approx \tau_1$. Meanwhile, we do not need to measure the control torques.

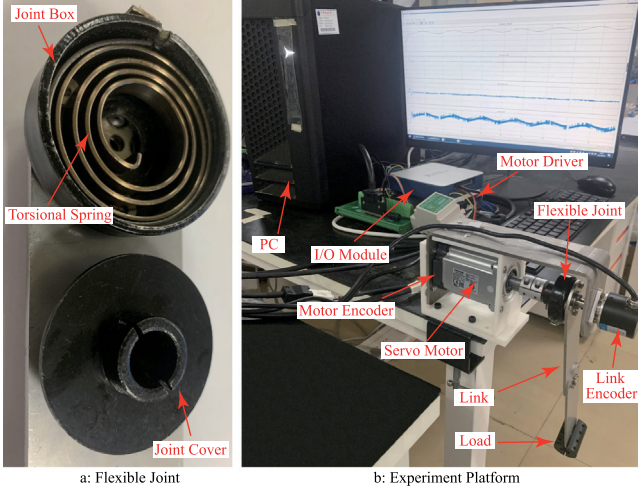


Fig. 6. Flexible joint and experiment platform of uncertain single-link FJM.

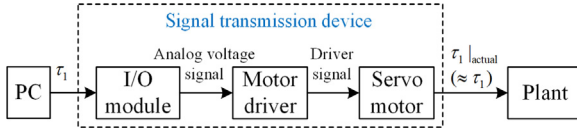


Fig. 7. Block diagram of torque signal transmission.

We choose the design parameters of the TVDO-based ISMC method as $l_{i11} = 125$, $l_{i21} = 75$, $l_{i31} = 15$, $c_{11} = 500$, $c_{21} = 90$, $c_{31} = 16$, $\lambda_{B1} = 5$, $\alpha_0 = 1$, and $\varepsilon = 0.5$. The design parameters $h_i(t)$ and $\lambda_{Ar}(t)$ are chosen to satisfy (24) and (30) respectively, where $t_1 = 3$ s, $v_{H1} = 2$, $v_{H2} = 1$, $t_2 = 6$ s, $v_{L1} = 5$, $v_{L2} = 10$, and $v_{L3} = 1$. Meanwhile, we also consider a special case by choosing constant design parameters for the proposed method, i.e., setting $h_i(t) = 1$, $\lambda_{Ar}(t) = 10$, and $S_0(t) = 0$.

In fact, the actual model of the FJM has unknown and complex uncertainties. Since the traditional SMC method and the proportional integral derivative (PID) method are capable of controlling the uncertain underactuated systems, we compare our proposed method with them in the experiment. For the traditional SMC method, τ_1 is designed as $\tau_1 = \tau_s := -g^{-1}[f(e_1) + c_{11}e_{21} + c_{21}e_{31} + c_{31}e_{41} + \lambda_{C1}S_C + \lambda_{C2}\text{sat}(S_C)]$, where $S_C := c_{11}e_{11} + c_{21}e_{21} + c_{31}e_{31} + e_{41}$. For the PID method, τ_1 is designed as $\tau_1 = \tau_p := -k_p e_{11} - k_i e_{11} - k_d e_{21}$, where k_p , k_i , and k_d are positive constants. To make the comparison as fair as possible, we choose some design parameters of the traditional SMC method as $c_{11} = 500$, $c_{21} = 90$, and $c_{31} = 16$, which are the same as those of the TVDO-based ISMC method in this experiment. Other design parameters of the traditional SMC method and the PID method are chosen after several experimental attempts to make the tracking error as small as possible. After many experiments, we select a set of optimal parameters that $\lambda_{C1} = 100$, $\lambda_{C2} = 30$, $\varepsilon = 0.5$, $k_p = 2$, $k_i = 1.6$, and $k_d = 1.5$.

The experimental results are shown in Fig. 8. It is clear that the TVDO-based ISMC method is effective in realizing tracking control of the UFJM. Meantime, the proposed method has better steady-state performance in tracking accuracy than the traditional SMC method and the PID method, and has better transient performance in motor torque than the special case “Constant” of the proposed method. It is worth mentioning that the experimental results also illustrate that the proposed method does not bring excessive computational pressure to the PC.

To further verify the superiority of the proposed method in controlling the actual FJM, we compare it with the DO-based SMC method (Ginoya et al., 2014). The torque of the DO-based SMC method is designed

as $\tau_1 = \tau_D := (e_{11} - e_{31})K_1 - B_{A1}e_{42} - I_{A1}[\hat{d}_4 + c_{11}e_{21} + c_{21}(e_{31} + \hat{d}_2) + c_{31}e_{41} + k_l S_D + k_s \text{sat}(S_D)]$, where $S_D := c_{11}e_{11} + c_{21}e_{21} + c_{31}e_{31} + e_{41} + c_{31}\hat{d}_2 + \hat{d}_2$, and $\hat{d}_2$ and $\hat{d}_4$ are the estimated values of σ_1 and σ_2 , obtained by a DO. This DO is similar to our TVDO with constant design parameters. For fair comparison, we choose the same values for the same type of design parameters in the proposed method and DO-based SMC method as far as possible, i.e., $c_{11} = 100$, $c_{21} = 60$, $c_{31} = 12$, $\lambda_{B1} = k_s = 2$, $\alpha_0 = \alpha_1 = 1$, $\varepsilon = 0.3$, $l_{i11} = 150$, $l_{i21} = 60$, $l_{i31} = 8$, and $k_l = 40$. We follow (24) and (30) to choose the time-varying $h_i(t)$ and $\lambda_{Ar}(t)$, in which $t_1 = 3$ s, $v_{H1} = 2$, $v_{H2} = 1$, $t_2 = 5$ s, $v_{L1} = 5$, $v_{L2} = 40$, and $v_{L3} = 1.5$. Here, we make the link angle track the target trajectory that

$$q_{1d} = \begin{cases} 0.15 \sin(0.8t - 0.5\pi) + 0.15, & t \leq 1.25\pi \text{ s} \\ 0.05 \sin(0.2t - 0.75\pi) + 0.35, & t > 1.25\pi \text{ s}. \end{cases} \quad (51)$$

The experimental results are shown in Fig. 9. Although both methods can achieve the tracking control of the UFJM, the DO-based SMC method exhibits significant torque chattering near the 2nd second. Therefore, the proposed method has better transient control performance than the DO-based SMC method.

6. Conclusion

In this paper, we tackled the tracking control problem of the UFJM via a TVDO-based ISMC method. The TVDO estimated disturbances and avoided the initial peaking phenomenon of observed states. The TVDO-based ISMC law overcame matched and mismatched disturbances, reduced chattering, and alleviated the contradiction between high tracking accuracy and suitable torque without peaking phenomenon. By simulation and experiment, the effectiveness of the proposed method for the UFJM was verified. Compared with the traditional SMC method, the PID method, the DO-based SMC strategy (Ginoya et al., 2014), the EID-based method (Yan et al., 2021), the NDO-based DSC method with IT2NNA (Hu et al., 2021), and the AFCF method (Ling et al., 2020), the proposed method showed better tracking control performance. In addition, the proposed method can also be extended to some other nonlinear systems with matched disturbances or mismatched disturbances.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Figs. 10 and 11.

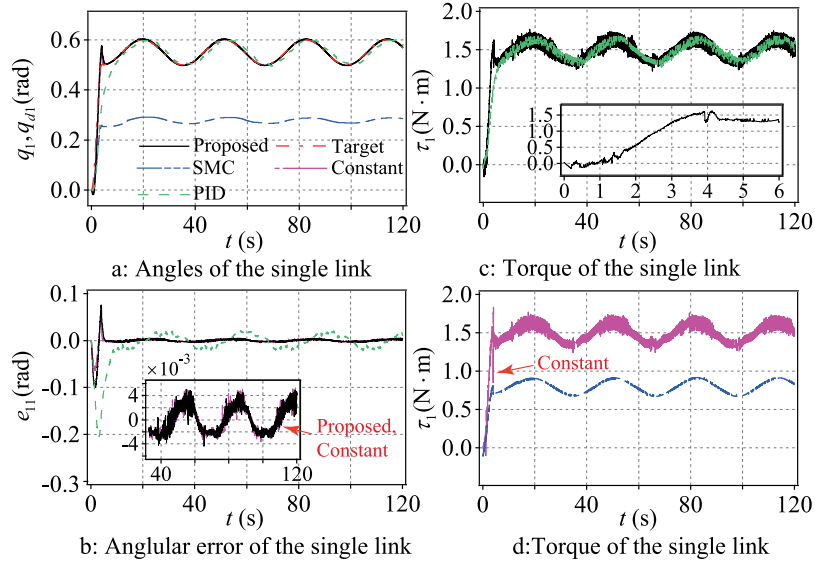


Fig. 8. Experimental results for UFJM with a single link (“Proposed”, “SMC” and “PID” employ the proposed method, traditional SMC method and PID method, respectively; “Target” denotes the target trajectory; “Constant” denotes the special case of the “Proposed” using constant design parameters).

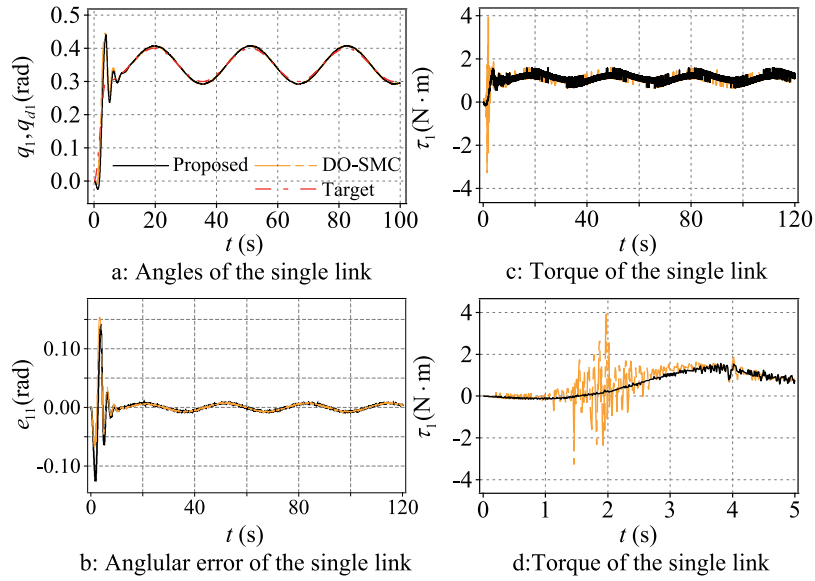


Fig. 9. Experimental results for UFJM with single-link (“Proposed”, “DO-SMC”, and “Target” denote the proposed method, the DO-SMC method, and target trajectory, respectively; The sub-figure d is an enlarged version of sub-figure c from 0 to 5 s.).

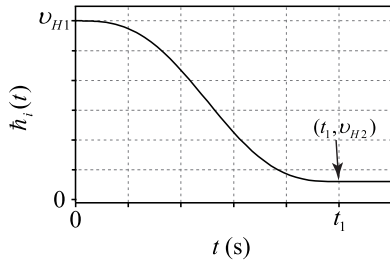


Fig. 10. Trajectory of $h_i(t)$.

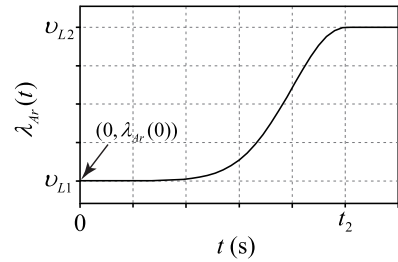


Fig. 11. Trajectory of $\lambda_{Ar}(t)$.

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